Data Product Generator for SAP Business Data Cloud

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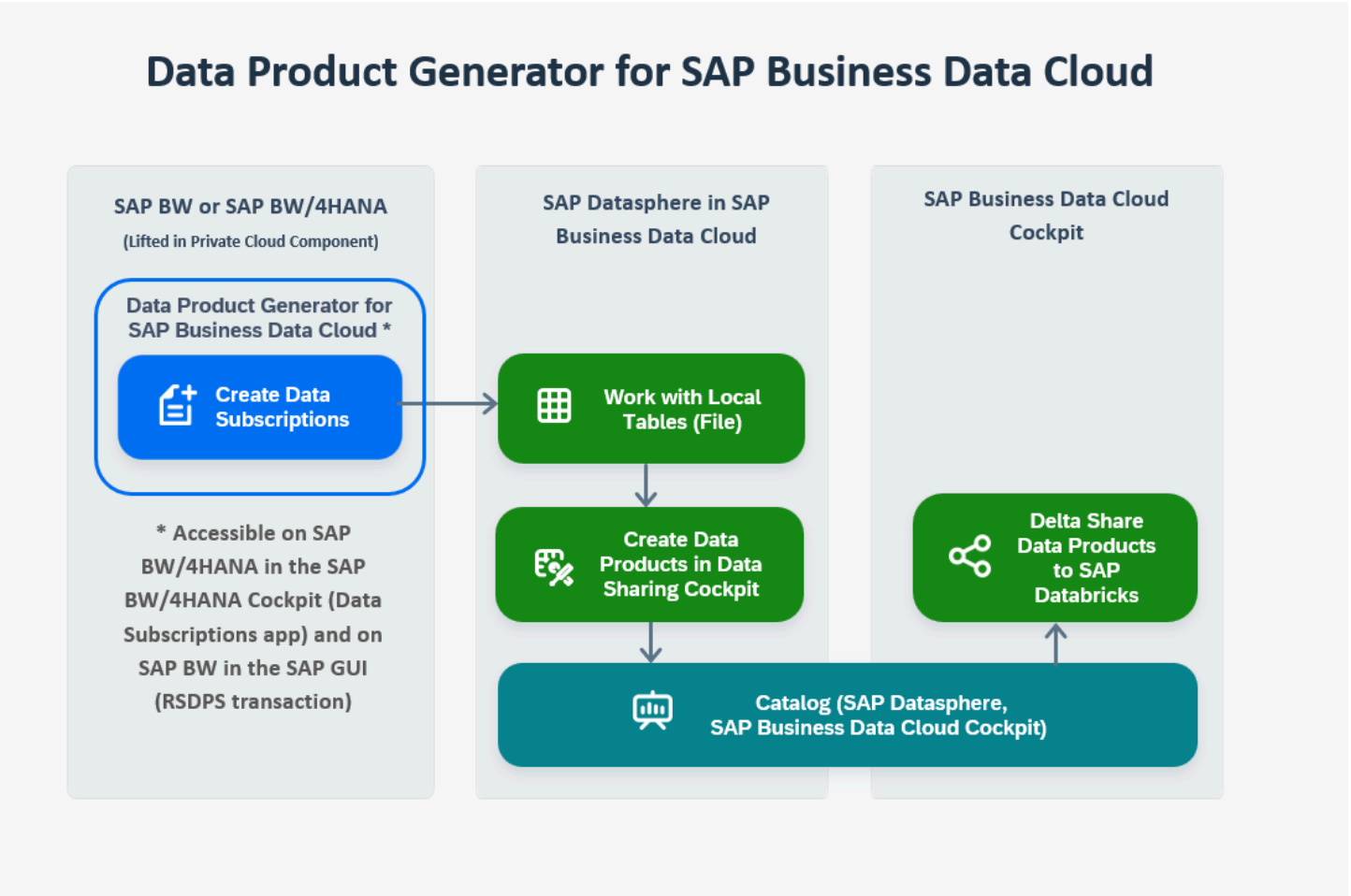
Data Product Generator for SAP Business Data Cloud

The Data Product Generator for SAP Business Data Cloud is a dedicated toolset available for SAP BW and SAP BW/4HANA allowing you to extract data from your existing BW InfoProviders into local tables (file) in the object store of SAP Datasphere in SAP Business Data Cloud.

To support the Data Product Generator, the data subscriptions have been introduced as a new object type in BW. You can create, maintain, and run the data subscriptions in a dedicated Data Subscriptions app in the SAP BW/4HANA Cockpit, or in the RSDPS transaction in the SAP GUI on older BW systems.

By creating a data subscription, you can replicate data from a specific BW InfoProvider into a local table (file). The local table (file) with the BW data will be generated into a dedicated BW file space which has been created during the initial configuration of the Data Product Generator. The local tables (file) in this space are read-only, but it's possible to perform data management tasks such as previewing, exporting, or deleting the data. To use the data in modeling scenarios, you can share it to other spaces where it can be consumed in views and analytic models.

To enable further use cases for SAP Business Data Cloud, you can create a data product out of the local table (file) in the Data Sharing Cockpit. The created data product can be accessed from the catalog in the SAP Business Data Cloud cockpit, where it can be delta shared to SAP Databricks for advanced AI/ML workflows on top of your BW data.



Disclaimer

In order to use the Data Product Generator for SAP Business Data Cloud, you need to be on a certain version of your BW system, lift your BW system into the private cloud component of SAP Business Data Cloud, and have an SAP Business Data Cloud cockpit tenant with an SAP Datasphere system in the formation.

To use more advanced functionalities, such as sharing data products to SAP Databricks, you might require further modules of SAP Business Data Cloud, such as the SAP Databricks capability, enabled within your SAP Business Data Cloud formation.

This documentation applies if you're on one of the following BW systems:

- SAP BW 7.5 SP24 or higher (SAP Business Warehouse - Help Portal Documentation)
- SAP BW/4HANA 2021 SP04 or higher (SAP BW/4HANA 2021 - Help Portal Documentation)
- SAP BW/4HANA 2023 SP00 or higher (SAP BW/4HANA 2023 - Help Portal Documentation)

Please note, that you can only use functions in the SAP BW/4HANA Cockpit, such as the Data Subscriptions app, on SAP BW/4HANA systems. Equivalent functions are offered within the RSDPS transaction in the SAP GUI on SAP BW 7.5 systems.

Some features of the Data Product Generator might only be available on higher support packages than the minimum system requirements for the Data Product Generator itself. For detailed information on the specific features and their required support packages, see What's New (Release Information).

What's New (Release Information)

The Data Product Generator for SAP Business Data Cloud is released via TCIs (Transport-Based Correction Instructions) that you can implement into your SAP BW or SAP BW/4HANA system as a SAP Note.

Release	What's New
Initial release (April 2025)	**Data Product Generator for SAP Business Data Cloud** The Data Product Generator for SAP Business Data Cloud is now available for SAP BW 7.5 SP24+, SAP BW/4HANA 2021 SP04+, and SAP BW/4HANA 2023 SP00+. With the Data Product Generator you can lift your on-premise BW system into the private cloud component of SAP Business Data Cloud and reuse existing BW data and objects by extracting them into local tables (file), creating data products, or enabling further use cases such as sharing to SAP Databricks. For more information, see Configuring the Data Product Generator or SAP Note 3590400. **Data Subscriptions** To support the Data Product Generator, the data subscriptions have been introduced as a new BW object type, allowing to save information about the dataset to be extracted from the selected InfoProvider and the extraction operation to be performed (e.g. selection, projection, extraction mode, ...). You can create, maintain, and run data subscriptions from the dedicated Data Subscriptions app in the SAP BW/4HANA Cockpit or through the RSDPS transaction in the SAP GUI. For more information, see Data Subscriptions.
New features and enhancements (December 2025) (SAP BW 7.5 SP24+, SAP BW/4HANA 2021 SP04+, SAP BW/4HANA 2023 SP04+)	**Multiple Targets for the Data Product Generator** When creating a new data subscription it is now possible to configure multiple targets for the Data Product Generator. The targets can be either different SAP Datasphere tenants or just different spaces in the same SAP Datasphere tenant. This allows to separate extracted BW data into different in SAP Datasphere tenants or spaces depending on functional area (e.g. FI, HCM). The targets can be maintained in the RSDPS transaction in the SAP GUI. For more information, see Configuring Targets for the Data Product Generator and Creating Data Subscriptions.

Release	What's New
	Mass Deletion of Data Subscriptions by Target With a new ABAP report available in the SAP GUI it is now possible to delete multiple data subscriptions at once if they have the same target. This allows for easier mass deletion of data subscriptions without having to delete each subscription manually. The functionality can also be accessed through the RSDPS transaction in the SAP GUI. With a force deletion option you can still delete the data subscriptions even if the SAP Datasphere tenant isn't available anymore. With a simulation mode you can preview the results of the deletion operation without having to perform any changes in production. For more information, see Mass Deleting Data Subscriptions by Target.
	Transport of Data Subscriptions With a new TLOGO type (DSUB) for the data subscriptions it is now possible to transport them between different BW systems (production, development, quality). The DSUBs support an A (Active) and an M (Modified) version. You can activate the data subscriptions in the RSDPS transaction in the SAP GUI or in the Data Subscriptions app in the SAP BW/4HANA Cockpit to prepare them for transporting. Here, you can also configure the transport request and package. In addition, there is a dedicated ABAP report available to migrate the data subscriptions from the old SUBS type to the new DSUB type. All new data subscriptions that are created are now automatically DSUBs. This enhancement also supports the integration of the data subscriptions into the BW Modeling Tools (BWMT). For more information, see Transporting and Migrating Data Subscriptions.
	Running Task Chains in Process Chains With a new process type for task chains it is now possible to include task chains from SAP Datasphere in process chains from SAP BW or SAP BW/4HANA. The necessary configurations can be made in the Process Chain Editor in the SAP BW/4HANA Cockpit after the task chain has been created in the Task Chain Editor in SAP Datasphere. This allows to schedule and run end-to-end processes, such as executing a transformation flow in a task chain after the data has been extracted via the data subscription. For more information, see Running Task Chains in Process Chains.
	Integration into BW Modeling Tools (BWMT) With the release of the BWMT 1.27, the Data Product Generator is now also integrated into the BW Modeling Tools, providing you with an Eclipse-based modeling interface to easily mass maintain data subscriptions. This includes the Scenario Generator, a new Eclipse view that enables mass creation of data subscriptions while considering object dependencies and existing data subscriptions (preventing the creation of duplicates). Through this you can import semantics and associated information from the BW InfoProviders into your SAP Business Data Cloud landscape, allowing you to reuse existing BW models for new use cases in SAP Business Data Cloud. For more information, see BW Data Product Generator Perspective in BW Modeling Tools (BWMT) and Using Scenario Generator.
	Technical User for Data Product Generator It's now possible to use a technical user to push data from the BW system to the object store in SAP Datasphere and to merge the delta data into the local tables (file). This removes the dependency on a business user for the connection to SAP Datasphere and allows for more stable operations, especially for scheduled and automated data push and merge scenarios, while not requiring personal user authorizations or account maintenance anymore. The necessary setup can be made during initial configuration of the Data Product Generator.

Release	What's New
	For more information, see SAP Note 3590400.
Enhancements and upgrades for BW Modeling Tools (February 2026) (BWMT 1.27.1)	**BW Data Product Generator Perspective for BW Modeling Tools** In addition to the two Eclipse views in the BW Modeling Tools, you can now also access all Data Product Generator functionalities via the dedicated BW Data Product Generator perspective in the BW Modeling Tools. The perspective provides a welcome screen with more information about the Data Product Generator and instant access to the tabs of the Scenario Generator and Data Subscriptions views. For more information, see BW Data Product Generator Perspective in BW Modeling Tools (BWMT).
	Scenario Generator Enhancements Enhancements for the Scenario Generator improve usability and provide further functions: • Target check for scenario: Via a new option when creating a scenario it's now possible to check the connection of a target before selecting it for the scenario. This makes it easier to avoid connection errors based on falsely configured targets when the scenario is executed. • Replacing multiple data subscriptions at once: Via a new dialog in the list of to-be-generated data subscriptions it's now possible to replace multiple data subscriptions at once with selected other ones based on certain criteria. This makes the multi-replace function easier to access so that you don't have to manually select data subscriptions in the list anymore. For more information, see Using Scenario Generator.
Enhancements and upgrades for BW Modeling Tools (March 2026) (BWMT 1.27.2)	**Delta Extraction for MultiProviders (MPRO) and CompositeProviders (HCPR)** It's now also possible to extract data from MultiProviders (MPROs) and CompositeProviders (HCPRs) via delta extraction with the Data Product Generator. Delta extraction is enabled for MPROs containing only InfoCubes and for HCPRs which are based exclusively on InfoCubes, Standard DataStore objects (advanced) (aDSOs), or Cube-Like DataStore objects (advanced) (aDSOs). Delta extraction from CompositeProviders is not supported if the CompositeProvider contains a mix of different objects, such as an InfoCube and a DataStore object (advanced), or if it contains any SAP HANA Views. For more information, see Setting Extraction Mode for the Data Product Generator or SAP Note 3695454.

Configuring the Data Product Generator

Configure the Data Product Generator for SAP Business Data Cloud to connect your on-premise SAP BW or SAP BW/4HANA system with your SAP BDC system to enable reuse of existing data in BW InfoProviders for scenarios across the SAP BDC landscape.

Preparing your SAP BW or SAP BW/4HANA System

Ensuring Minimum Systems Requirements

Before you can use the Data Product Generator, ensure you're on a supported SAP BW or SAP BW/4HANA system which has been lifted into the private cloud component of SAP Business Data Cloud. In addition, you need to have an SAP Business Data Cloud cockpit tenant with an SAP Datasphere system in the formation.

The Data Product Generator for SAP Business Data Cloud is available for SAP BW 7.5 SP24 or higher, SAP BW/4HANA 2021 SP04 or higher, and SAP BW/4HANA 2023 SP00 or higher.

i Note

The Data Product Generator is **not** available for embedded SAP BW systems, such as in SAP S/4HANA. (See, SAP Note 3584640.)

i Note

Some features of the Data Product Generator are only available on higher support packages than the minimum systems requirements for the Data Product Generator itself. For more information, see What's New (Release Information).

Your SAP BW or SAP BW/4HANA system needs to be in a private cloud environment in SAP Business Data Cloud, also referred to as SAP Business Warehouse, private cloud edition for SAP NetWeaver, or SAP BW/4HANA, private cloud edition. If your SAP BW or SAP BW/4HANA system isn't in a private cloud environment yet, please get in touch with SAP's sales representatives to assist you with lifting your BW environment into the private cloud component of SAP BDC.

You also need to have an SAP Business Data Cloud cockpit tenant with an SAP Datasphere system in the formation. For further use cases, such as sharing to SAP Databricks, you would require further components of SAP Business Data Cloud, such as the SAP Databricks capability.

→ Tip

It is possible to rewire your existing standalone SAP Datasphere tenant into an SAP Business Data Cloud system. For more information, see Updating Your SAP Datasphere and SAP Analytics Cloud Subscriptions to SAP Business Data Cloud.

Configuring the Data Product Generator

Enabling Object Store for SAP Datasphere

The object store needs to be enabled and configured in your SAP Datasphere tenant.

To configure the object store in SAP Datasphere, go to **Configuration** > **Tenant Configuration** > **Object Store**. Here you can make the necessary settings for **Storage** (storage size of object store in TB), **Compute** (computing power in block-hours), and **Requests** (number of API calls needed per month). For more information, see Configure the Size of Your SAP Datasphere Tenant.

Configuration of the Data Product Generator

To configure the Data Product Generator, follow the steps provided in SAP Note 3590400.

You need to conduct the configurations for the first-time setup of the Data Product Generator manually.

To get started with the process, open a support case on component BDC-BW.

i Note

This is needed to authorize you to use the Data Product Generator and register your BW system on the SAP Business Data Cloud cockpit tenant with the SAP Datasphere system in the formation. However, you still need to configure the Data Product Generator manually by yourself.

If your systems and existing licenses fulfill the requirements, the necessary instructions and tooling for the configuration will be provided to you.

For the manual configurations, refer to the **Configuring the Data Product Generator** guide which you can find here: Configuring the Data Product Generator for SAP Business Data Cloud.

Configuring Targets for the Data Product Generator

You can configure multiple targets for the Data Product Generator, which allows you to separate extracted BW data into different BW file spaces or SAP Datasphere tenants based on functional area (e.g. FI, HCM)). The target for a data subscription can be selected if you create the data subscription.

Viewing and Maintaining Targets

You can view and maintain all targets for your BW system in the RSDPS transaction in the SAP GUI.

If you navigate to Goto > Maintain Target System, a new screen will open that lets you view all existing targets, as well as create new ones, or edit and delete existing ones.

In the Display Target System screen, you can perform the following actions for targets:

Actions for Targets

Action	Procedure
View target details	To view the properties of any existing target in your BW system, select the target in the Target Name field. As a result, the fields will be filled in with the information from the selected target.
Check connectivity for target	Select Check Connectivity ( ) to check the connection to SAP Datasphere for the displayed target.
Edit target	Select Display ↔ Change ( ) to open the edit mode for the displayed target. Change the settings you want to adjust, then select Save ( ) to confirm.
Delete target	Select Display ↔ Change ( ) and then Delete Target System ( ) to delete the displayed target. You need to confirm the deletion in an additional pop-up window.
Create new target	Select Display ↔ Change ( ) and then Create Target System ( ) to add new targets.

Creating New Targets

You can create a new target for the Data Product Generator in the RSDPS transaction in the SAP GUI.

1. In the top bar of the RSDPS transaction, select Goto > Maintain Target System.
2. Select Display ↔ Change ( ) and then Create Target System ( ).
3. Enter a Target Name in the pop-up window.
4. Input further information about the new target as required.
    a. Set the Target Type to HDLFS (SAP HANA Cloud, data lake Files) as this is the only target type supported by the Data Product Generator.

b. Tick off the checkbox **Preferred System** if you want to set the new target as the user-preferred target for your BW system.

c. Enter a **Description** for your target.

d. Enter details about the connection to SAP Datasphere for the new target. For more information, see SAP Note 3590400.

5. (Optional) Select **Check Connectivity** ( ) to test the connection to SAP Datasphere and resolve connection errors as required.

6. Select **Save** ( ) to save the new target.

   You can now use this target if you create new data subscriptions.

Note

For each new target you need to implement the configuration task list for the Data Product Generator which includes the setup of a dedicated BW file space in the SAP Datasphere tenant to receive pushed data. For the full procedure, see SAP Note 3590400.

After the targets have been configured they can be selected if you create a new data subscription, as well as viewed in the data subscriptions list.

Selecting Targets for Data Subscriptions

You can select the target for a data subscription when you create it:

- **Creating a data subscription:**
  - **Data Subscriptions app (SAP BW/4HANA Cockpit):** If you create a data subscription in the SAP BW/4HANA Cockpit, you can set the target in the **Target Name** field. If you toggle the arrow icon, you can choose from all existing targets in your BW system. The value help will display their names and descriptions. (For more information, see Creating Data Subscriptions.)
  - **RSDPS transaction (SAP GUI):** If you create a data subscription in the SAP GUI, you can set the target in the **Target Name** field. If you click on the file icon, you can choose from all existing targets in your BW system. The value help will display their names, descriptions, and whether they're a preferred target (marked with an X). The **Target Type** field will be pre-filled with HDLFS (SAP HANA Cloud, data lake Files) since this is the only available target type for the Data Product Generator. (For more information, see Creating Data Subscriptions.)

    ### Note

    If a user-preferred target has been configured, the **Target Name** field will be initially filled with this target. If no user-preferred target has been configured, it will be filled with the system-preferred target. The **Target Name** field cannot be left empty or filled with invalid input (such as a target that doesn't exist), otherwise the data subscription cannot be saved.

    ### Caution

    Once the data subscription has been created, it's not possible to adjust the target anymore. Therefore, make sure to select the correct target during first-time creation.

- **Creating a data subscription from template:** If you create a data subscription from template, the target will be taken over from the template data subscription. However, before saving the new data subscription, you can still select a different target. This allows you to copy existing data subscriptions but enable them for different targets. (For more information, see Creating Data Subscriptions from Template.)

You can also select the target for a scenario in the Scenario Generator:

- **Creating a scenario:** In the **Scenario Core Objects** tab of the Scenario Generator view you can pre-define the target for the entire scenario in the **Target Name** field. Select **Browse** to choose from all existing targets in your BW system. This target will be applied to all to-be-generated data subscriptions of the scenario which will be visible in the **Objects Relations** tab under **Selection to Generate Data Subscriptions**. In the second step you can here still adjust the targets of the individual data subscriptions. (For more information, see Creating Scenarios.)
- **Generating data subscriptions**: Before you generate the data subscriptions out of the scenario tree, you still can change the target for the individual data subscriptions in the scenario. To do this, go to the **Object Relations** tab of the Scenario Generator view. In the section **Selection to Generate Data Subscriptions**, select the data subscriptions for which you want to set a different target, then right-click and select **Edit**. Go to the field **Target Name** and select **Browse** to choose from all existing targets on your BW system. (For more information, see Generating Data Subscriptions.)

  ⚠ **Caution**

  Generally, you can use multiple targets for the Data Product Generator in the BW Modeling Tools. However, all data subscriptions in a scenario must belong to the same target to enable the data subscriptions to be generated. It's possible to have multiple different targets for different scenarios. For this reason, the target is best to be defined on scenario-level in the Scenario Generator, and not for the individual data subscriptions.

Viewing Targets for Data Subscriptions

You can view the targets of the data subscriptions in the **Target Name** column, which is visible in the various different interfaces of the Data Product Generator.

Interface	Viewing Targets
RSDPS transaction (SAP GUI)	• **Target Name** column in the data subscriptions list. • **Target Name** and **Target Type** fields in the **Details** screen of a data subscription. • **Target Name** and **Target Type** fields in the **Edit Subscription** screen. (For more information, see Editing Data Subscriptions.)
Data Subscriptions app (SAP BW/4HANA Cockpit)	• **Target Name** column in the data subscriptions list. • **Target Name** field in the **Maintain Subscription** screen. (For more information, see Editing Data Subscriptions.) ℹ **Note** The **Target Name** column must first be toggled via **Settings** (⚙) to make it visible in the data subscriptions list. This is different than in the SAP GUI and in the BW Modeling Tools, where it is visible by default. In the SAP BW/4HANA Cockpit it's also possible to filter the data subscriptions list by **Target Name** via the respective filter settings.

Interface	Viewing Targets
Data Subscriptions view (BW Modeling Tools)	• **Target Name** column in the data subscriptions list. (For more information, see BW Data Product Generator Perspective in BW Modeling Tools (BWMT).)
Scenario Generator view (BW Modeling Tools)	• **Target Name** column under **Collected Objects** in the **Object Relations** tab. Here you can see the targets for existing data subscriptions associated with the selected BW models and its dependencies. (For more information, see Creating Scenarios.) • **Target Name** column under **Selection to Generate Data Subscriptions** in the **Object Relations** tab. Here you can see the targets you selected individually for the data subscriptions to be generated for the scenario. (For more information, see Generating Data Subscriptions.)

Setting Extraction Mode for the Data Product Generator

With the Data Product Generator you can run a full extraction or a delta extraction to the local table (file). Depending on the InfoProvider you're extracting from, capabilities for delta extraction are available. You can make the necessary settings in the data subscription.

Extraction Modes for the Data Product Generator

There are two types of extraction modes available for the Data Product Generator:

- **Full extraction (Full):** Extracts all data from the InfoProvider at once (also called snapshot) in a one-time operation.
- **Delta extraction (Delta):** Extracts after initial extraction only delta data from the InfoProvider.

In both modes the extracted data arrives in the inbound buffer of the object store in SAP Datasphere and you need to schedule and run a merge task (either manually or via task chain) to merge the new snapshot data or delta data into the local table (file). (For more information, see Viewing Local Table (File) and Getting Data Updates).

Availability of Delta Extraction based on BW System and InfoProvider

Delta extraction is only available for specific InfoProviders based on the BW system.

⚠ Caution

Before you try to configure delta extraction for your data subscription, check the type of InfoProvider you want to extract data from and the BW system you're on.

Delta Extraction for InfoProviders in SAP BW/4HANA and SAP BW 7.5

InfoProvider	SAP BW/4HANA	SAP BW 7.5
Standard DataStore object (advanced) (aDSO)	Yes	Yes

InfoProvider	SAP BW/4HANA	SAP BW 7.5
DataStore object (advanced) (aDSO) of type data mart	Yes	Yes
Staging DataStore object (advanced) (aDSOs) with properties Inbound Queue Only, Compress Data, or Reporting-Enabled	Yes	Yes
InfoObjects with Enhanced Master Data Update	Yes	n/a
Standard and Write-Optimized DataStore object (classic) (ODSO)	n/a	Yes
InfoCube	n/a	Yes
CompositeProvider (HCPR) *	Yes	Yes
MultiProvider (MPRO) *	No	Yes

ℹ Note

(*) The following restrictions apply to delta extraction capabilities of HCPRs and MPROs:

- Delta extraction from MultiProviders (MPROs) is only supported for MultiProviders that only contain InfoCubes. For MultiProviders containing other InfoProviders (such as DataStore objects) or a mix of InfoCubes and other InfoProviders only full extraction is possible.
- Delta extraction from CompositeProviders (HCPRs) is only supported for CompositeProviders that only contain InfoCubes, Standard DataStore objects (advanced) (aDSOs) or Cube-Like DataStore objects (advanced) (aDSOs). In addition, all contained objects must be of the same type, for example several InfoCubes. It's not possible to use delta extraction for CompositeProviders containing a mix of different objects, such as an InfoCube and a DataStore object (advanced).
- Delta extraction from CompositeProviders (HCPRs) is not supported if the CompositeProvider contains SAP HANA Views.

Extracting Archived Data from BW Systems

The Data Product Generator can also extract data directly from NLS (Near-Line Storage) into the local table (files), which applies to both SAP BW 7.5 and SAP BW/4HANA. This, however, is not possible for data archived with the ADK-based Data Archiving Process (DAP), which only applies to SAP BW 7.5.

NLS allows you to offload historical or rarely accessed data from SAP HANA, the primary database used for BW, into a secondary database like SAP IQ while keeping the data accessible for reporting and analysis. It is the modern data archiving solution available in SAP BW 7.5 and SAP BW/4HANA.

The classic data archiving in SAP BW 7.5 works through the ADK-based Data Archiving Process (DAP) and stores archived data outside the BW system. Here, the archived data is not queryable anymore and therefore not available for reporting and analysis (or extraction with the Data Product Generator). NLS also uses DAP as an archiving mechanism, however not the ADK-based DAP from SAP BW 7.5.

Setting Extraction Mode for Data Subscriptions

You can select the extraction mode for a data subscription when you create it:

- **Data Subscriptions app (SAP BW/4HANA Cockpit):** If you create a data subscription in the SAP BW/4HANA Cockpit, you can set the extraction mode in the **Extraction Mode** field. For this, you need to switch to the **Settings** tab in the **Maintain Subscription** screen. (For more information, see Creating Data Subscriptions).

- **RSDPS transaction (SAP GUI):** If you create a data subscription in the SAP GUI, you can set the extraction mode in the **Extraction Mode** field in the **General** section of the **Edit Subscription** screen. (For more information, see Creating Data Subscriptions).

ℹ Note

You can still change the extraction mode of a data subscription after its initial creation, but only if you haven't run it yet. After first run, the extraction mode will become read-only and cannot be changed anymore. (For more information, see Editing Data Subscriptions.)

You can also select the extraction mode for data subscriptions in a scenario:

- Scenario Generator view (BW Modeling Tools): If you have created a scenario and want to generate the associated data subscriptions, you can change the extraction mode of the individual data subscriptions in the **Selection to Generate Data Subscriptions** section of the **Object Relations** tab. Here, select the data subscriptions for which you want to set a different extraction mode, right-click and choose **Edit**, then set the extraction mode in the **Extraction Mode** field. (For more information, see Generating Data Subscriptions.)

ℹ Note

It's not possible to set the extraction mode on scenario-level. If you want to change the extraction mode from the default (full extraction), you need to edit it for the individual data subscriptions.

Data Subscriptions

Data subscriptions are a new BW object type that have been introduced to support the Data Product Generator in SAP BW and SAP BW/4HANA systems. For more information, see SAP Note 3593519.

There are three ways to maintain the data subscriptions:

- Data Subscriptions App in SAP BW/4HANA Cockpit (SAP BW/4HANA 2021 and SAP BW/HANA 2023 systems)
- RSDPS Transaction in SAP GUI (SAP BW/4HANA 2021, SAP BW/4HANA 2023 and SAP BW 7.5 systems)
- BW Data Product Generator Perspective in BW Modeling Tools (BWMT) (SAP BW/4HANA 2021, SAP BW/4HANA 2023 and SAP BW 7.5 systems)

ℹ Note

On SAP BW 7.5 systems you need to use the RSPDS transaction in the SAP GUI instead of the Data Subscriptions app in the SAP BW/4HANA Cockpit to maintain the data subscriptions. The BWMT integration is available on SAP BW 7.5 and on SAP BW/4HANA systems.

Data Subscription (DSUB and SUBS)

A data subscription is a new BW object type that allows you to extract data from ODP-based sources within your BW system into local tables (file) in the SAP Datasphere system of your SAP Business Data Cloud formation.

Type of Data Subscription

Data subscriptions can have two types:

- **DSUB** ( ) - This is a transportable data subscription which has either been directly created in the DSUB format or migrated from the SUBS format via a dedicated migration report. Data subscriptions of type DSUB can be transported between different systems (e.g. production, development) like any other TLOGO. With DSUB the data subscriptions also receive a dedicated TLOGO type (similar to ELEM for BW Queries or ADSO for DataStore objects (advanced)).
- **SUBS** ( ) - This is a local data subscription which cannot be transported between systems. Data subscriptions of type SUBS can be migrated to the TLOGO type DSUB via a dedicated migration report.

> **ℹ Note**
>
> All newly created data subscriptions will automatically be of the type DSUB. This means Creating Data Subscriptions and Creating Data Subscriptions from Template are now only working for the type DSUB. Data subscriptions of type SUBS can still be maintained via Editing Data Subscriptions.

Information in a Data Subscription

A data subscription contains the information which ODP-based source (InfoProvider) you want to extract data from, which fields and filters you have selected to define the dataset to be extracted, as well as which further settings you have made for the extraction operation (extraction mode, projection, ...).

It doesn't contain the dataset to be extracted itself, but rather the definition of the extraction operation to be performed on a selected source object.

Some of the information in the data subscription can be manually set while you create the data subscription (for more information, see Creating Data Subscriptions). Others, such as the technical name and details about the local table (file) that will be generated during the extraction operation, will be automatically filled in.

Properties of a Data Subscription

The information in a data subscription is saved as its properties, which can be accessed in different ways:

- In the RSDPS transaction by opening the details screen of a data subscription via **Details** ( ) or by opening its edit screen via **Edit** (Edit). In the edit screen, you can not only view all properties in a comprehensive format, but also change the editable ones.
- In the Data Subscriptions app by clicking on any of the data subscriptions in the list. This opens the **Maintain Subscription** screen, where you can view a select amount of properties, as well as change the editable ones.

You can also view some of the properties as columns in the data subscriptions list, as well as filter or search the list based on certain fields and values. For more information, see Data Subscriptions App in SAP BW/4HANA Cockpit, RSDPS Transaction in SAP GUI or BW Data Product Generator Perspective in BW Modeling Tools (BWMT).

Properties of a Data Subscription

Field	Information
Technical Name	Technical name of the data subscription which is generated automatically during its creation
Description	Description of the data subscription which can be manually entered during its creation

Field	Information
Source Name	Selected source for the data subscription (InfoProvider from the BW system)
Source Type	Type of the selected source for the data subscription (Source type for the Data Product Generator is always ODP (Operational Data Provisioning))
Target Name	Selected target for the data subscription (Space/Tenant in SAP Datasphere)
Target Type	Type of the selected target for the data subscription (Target type for the Data Product Generator is always HDLFS (SAP HANA Cloud, data lake Files))
Subscription Type	Type of the data subscription (DSUB/SUBS)
Status	Status of the data subscription (Active/Inactive)
Extraction Mode	Extraction mode for the data subscription (Full/Delta)
Scheduling	Scheduling settings for the data subscription (None/Direct Execution or Process Chain)
Last Run	Information on whether the data subscription has been run yet and whether the last run has been successful or not
Last Changed At	User who has last changed the data subscription
Last Changed By	Date and time when the data subscription has been last changed (Timestamp)
Created At	Date and time when the data subscription has been created (Timestamp)
Created By	User who has created the data subscription

Data Subscriptions in the Catalog

If you have configured the metadata extraction from your BW system, you can see the data subscriptions amongst other assets and data products in the catalog. There you can view the details of the data subscription by opening the details screen.

⚠ Caution

The metadata extraction for data subscriptions is only available on SAP BW/4HANA 2023 SP04 or higher. This feature works independently from the Data Product Generator for SAP Business Data Cloud.

⚠ Caution

If you want to use this functionality, you first need to implement SAP Note 3601550 into your BW system and perform the manual steps at the end of the SAP Note. In addition, you need to fulfill all necessary prerequisites to connect a SAP BW/4HANA system to the catalog. For more information, see Checking the BW System Setup.

To set up the metadata extraction and connect your BW system, follow the detailed procedure in Creating a Connection to a BW System.

ℹ Note

While you need to set up the metadata extraction for your BW system in the SAP Datasphere tenant, the assets for the data subscriptions will be visible across all catalog entry points including SAP Datasphere (for modeler users and data analysts) and the SAP Business Data Cloud cockpit (for administrator users).

Data Subscriptions App in SAP BW/4HANA Cockpit

The Data Subscriptions app in the SAP BW/4HANA Cockpit provides BW administrators on a SAP BW/4HANA system with a modern and intuitive interface to view, manage, and maintain data subscriptions.

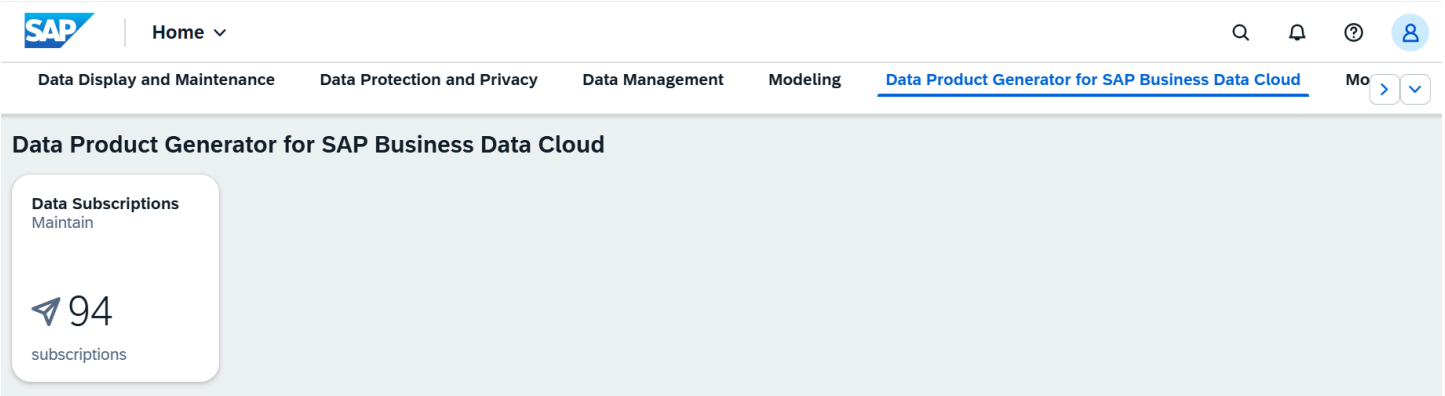
Accessing the Data Subscriptions App

You can access the **Data Subscriptions** app in the SAP BW/4HANA Cockpit under the tab **Data Product Generator for SAP Business Data Cloud**.

On the tile you can view the current number of data subscriptions in your BW system.

ℹ Note

On SAP BW 7.5 systems you need to use the RSPDS transaction in the SAP GUI to maintain the data subscriptions.



Data Subscriptions List - Columns and Values

The columns in the data subscriptions list display information about certain properties of the data subscriptions:

Columns in Data Subscriptions List

Column	Value
Subscription	Technical name and description of the data subscription.
Subscription Type	Type of the data subscription. • **DSUB** - Transportable data subscription (TLOGO type) • **SUBS** - Local data subscription (Cannot be transported)
Status	Shows whether the data subscription has been activated or not. • **Active, executable** - Data subscription has been activated and can be run.

Column	Value
	• **Inactive, not executable** - Data subscription still needs to be activated to be run.
Source Object	Name of the ODP-based source (InfoProvider) selected for the data subscription.
Source Type	Type of the selected source for the data subscription. Source type for the Data Product Generator is always ODP (Operational Data Provisioning).
Target Name	Name of the target selected for the data subscription.
Target Type	Type of the target selected for the data subscription. Target type for the Data Product Generator is always HDLFS (SAP HANA Cloud, data lake Files).
Last Changed At	Date and time at which the data subscription has last been changed.
Last Changed By	User who last changed the data subscription.
Created At	Date and time at which the data subscription has been created.
Created By	User who created the data subscription.
Owner	User who owns the data subscription.
Scheduling	Shows scheduling settings configured for data subscription. • **Immediate, Single Execution** - Data subscription has been scheduled via process chain. • **No Scheduling** - Data subscription needs to be run manually via the **Run** button.
Process Chain ID	ID of the process chain selected for running the data subscription. (Only applies if scheduling settings are set to process chain.)
Process Chain Description	Description of the process chain selected for running the data subscription. (Only applies if scheduling settings are set to process chain.)
Process Variant	Name of the process variant selected for running the data subscription. (Only applies if scheduling settings are set to process chain.)
Delta Runtime	Shows whether delta extraction has been configured for data subscription. • **No** - Full extraction has been set. • **Yes** - Delta extraction has been set.
Last Execution	Shows whether data subscription has already been run. • **Not executed** - Data subscription hasn't been run yet. • **Success** - Data subscription has been run yet and last run was successful.

Column	Value
	• Failed - Data subscription has been run yet, but last run has failed.
Local Table (File)	Click on View Local Table (File) to open the local table (file) in the SAP Datasphere tenant.

Data Subscriptions Toolbar - Configurations for Data Subscriptions List

In the search section above the data subscriptions list, you can search and filter the list:

- Via Adapt Filters you can add columns to the search section. For each of these columns you can specify filters for the list (e.g. only show data subscriptions of type SUBS). For more information on the columns that can be added, see section "Columns and Values".
- Via Easy Filter ( ) you can switch to a simplified search where you can enter search queries and filters in natural language.

You can also perform the following configurations on the list of data subscriptions:

Configurations for Data Subscriptions List

Button	Configuration
Refresh	Refresh the list of data subscriptions to show latest changes according to new sorting criteria or performed actions.
Copy to Clipboard ( )	Copy the data from the selected row in the list to the clipboard, allowing for easy copy/paste.
Show More per Row ( ) / Show Less per Row ( )	Expand or limit the information shown per row. This option is useful if you have configured many columns/properties to be shown for the data subscriptions and don't have enough space to view them all.
Settings ( )	Open the Settings menu to configure which columns/properties of the data subscriptions are shown in the list and in which order they appear. You can also define filters or groups. For more information on which columns can be shown, see section "Columns and Values".
Export Table ( )	Save the currently displayed list as a local file. If you select the dropdown besides the icon ( ) and then choose Export As, you can make further settings for the file format and file name, as well as whether you want to consider current filters while exporting or split cells with multiple values.

Data Subscriptions Toolbar - Actions for Data Subscriptions

You can perform the following actions on the data subscriptions:

Actions for Data Subscriptions

Button	Action
Run	Run the selected data subscription. This manually starts an extraction operation from the InfoProvider into the local table (file).

Button	Action
	Alternatively, you can also schedule the extraction operation via process chain.
Delete	Delete the selected data subscription. This doesn't delete the local table (file) if it has already been deployed. For more information, see Deleting Data Subscriptions.
Create from Template	Create a new data subscription based on a template of an existing data subscription. For more information, see Creating Data Subscriptions from Template.
Create	Create a new data subscription. For more information, see Creating Data Subscriptions.
Toggle Status	Activate or deactivate the selected data subscription. This toggles its status between **Active** and **Inactive**. A data subscription needs to be activated before you can run it. For more information, see Transporting and Migrating Data Subscriptions.

RSDPS Transaction in SAP GUI

The RSDPS transaction in the SAP GUI provides BW administrators on an SAP BW or SAP BW/4HANA system with an advanced interface to view, manage, and maintain data subscriptions. In the SAP GUI you can also access further functions to maintain targets and mass-delete data subscriptions based on target.

Accessing the RSDPS Transaction

You can access the RSDPS transaction in the SAP GUI after logging on to your BW system.

The main screen of the RSDPS transaction shows the list of all current data subscriptions in your BW system belonging to the Data Product Generator.

The list view can be filtered according to certain field values (e.g. only show data subscriptions of type DSUB).

Data Subscriptions List - Columns and Values

The columns in the data subscriptions list display information about certain properties of the data subscriptions:

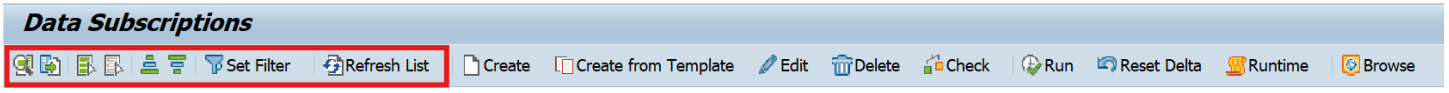
Columns in Data Subscriptions List

Column	Value
Selection	Shows whether data subscription is included in current selection in list.
Subscription	Technical name of the data subscription.

Column	Value
Type	Type of the data subscription. • DSUB ( ) - Transportable data subscription (TLOGO type) • SUBS ( ) - Local data subscription (Cannot be transported)
Long Description	Description of the data subscription.
Status	Shows whether the data subscription has been activated or not. • Active ( ) - Data subscription has been activated and can be run. • Inactive ( ) - Data subscription still needs to be activated to be run.
Revision	Shows whether A (Active) and M (Modified) versions of data subscription are equal. • Equal ( ) - A und M version of data subscription have no differences. The object is ready to be transported. • Not Equal ( ) - Data subscription has changes in M version that haven't been activated yet. The object should be activated before you try to transport it.
Source Name	Name of the ODP-based source (InfoProvider) selected for the data subscription.
Target Name	Name of the target selected for the data subscription.
Delta	Shows whether delta extraction has been configured for data subscription. • If field is empty, full extraction is set. • If field shows a checkbox ( ), delta extraction is set.
Owner	User who owns the data subscription.
Scheduling	Shows scheduling settings configured for data subscription. • Process Chain ( ) - Data subscription will be scheduled via process chain. • None/Direct Execution ( ) - Data subscription needs to be run manually via the **Run** button.
Last Run	Shows whether data subscription has already been run. • Not Run ( ) - Data subscription hasn't been run yet. • Last Run Successful ( ) - Data subscription has been run yet and last run was successful.

Column	Value
	• Last Run Failed ( ) - Data subscription has been run yet, but last run has failed.
Last Changed By	User who last changed the data subscription.
Last Changed At	Date and time at which the data subscription has last been changed.

RSDPS Toolbar - Configurations for Data Subscriptions List

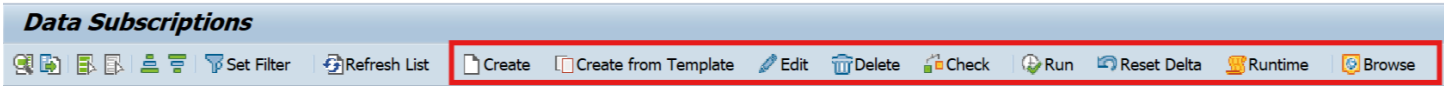


You can perform the following configurations on the list of the data subscriptions:

Configurations for Data Subscriptions List

Icon	Configuration
	Select a data subscription from the list and display its details screen. For more information on the shown properties, see Data Subscription (DSUB and SUBS).
	Save the currently displayed list in a local file (e.g. HTML, Rich Text, ...).
	Select all data subscriptions in the list.
	Deselect all data subscriptions in the list.
	Sort the data subscriptions in the list in ascending order. You can specify the fields for which you want to sort in a separate screen.
	Sort the data subscriptions in the list in descending order. You can specify the fields for which you want to sort in a separate screen.
Set Filter	Filter the list of data subscriptions according to certain fields and their values. For example, select the field "Type" and specify the value "SUBS" to only display data subscriptions from the old SUBS format that still need to be migrated.
Refresh List	Refresh the list of data subscriptions to show latest changes according to new sorting criteria or performed actions.

RSDPS Toolbar - Actions for Data Subscriptions



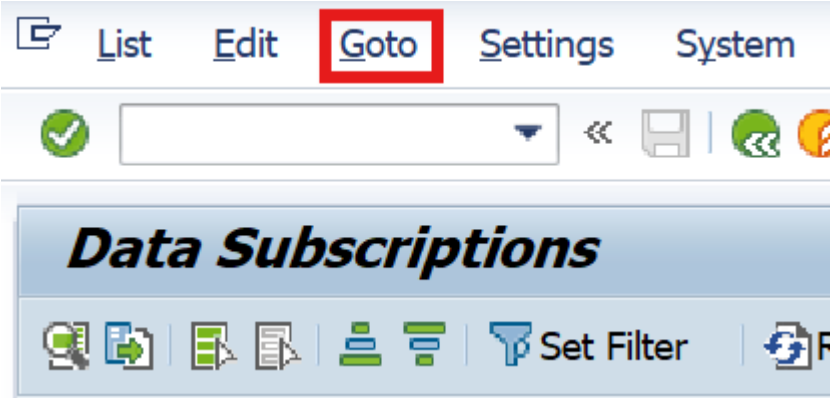
You can perform the following actions on the data subscriptions:

Actions for Data Subscriptions

Icon	Action

Icon	Action
Create	Create a new data subscription. For more information, see Creating Data Subscriptions.
Create from Template	Create a new data subscription based on a template of an existing data subscription. For more information, see Creating Data Subscriptions from Template.
Edit	Edit the selected data subscription. After the local table (file) has been created and deployed, options for editing become very limited. For more information, see Editing Data Subscriptions.
Delete	Delete the selected data subscription. This doesn't delete the local table (file) if it has already been deployed. For more information, see Deleting Data Subscriptions.
Check	Check the selected data subscription for any potential issues. In the results screen you can see any warnings or error messages concerning the status of the source (InfoProvider), the target, the connection to SAP Datasphere, and the local table (file).
Run	Run the selected data subscription. This manually starts an extraction operation from the InfoProvider into the local table (file). Alternatively, you can also schedule the extraction operation via process chain.
Reset Delta	Manually reset delta extractions for the selected data subscription. This is only relevant if you have delta extraction enabled as the extraction mode. For more information, see Setting Extraction Mode for the Data Product Generator.
Runtime	View more information about the last run for the selected data subscriptions. This is only possible for data subscriptions you have already run, otherwise you will receive an error message.
Browse	View the data in the local table (file) of the selected data subscription. For more information, see Viewing Local Table (File) and Getting Data Updates.

SAP GUI Toolbar - Access to other RSDPS Programs



In the topbar of the SAP GUI in the RSDPS transaction, you have access to further programs related to the Data Product Generator and the data subscriptions.

Select **Goto** and from there access:

- **Maintain Target System**: Lets you create new targets for the Data Product Generator, as well as view and maintain existing ones. For more information, see Configuring Targets for the Data Product Generator.
- **Mass-Delete Data Subscriptions by Target**: Lets you mass-delete all data subscriptions for one target. For more information, see Mass Deleting Data Subscriptions by Target.

BW Data Product Generator Perspective in BW Modeling Tools (BWMT)

The Scenario Generator in the BW Modeling Tools (BWMT) provides BW modelers on an SAP BW or SAP BW/4HANA system with a dedicated Eclipse view to mass generate data subscriptions based on object dependencies and existing data subscriptions. The Data Subscriptions view allows to view and mass maintain existing data subscriptions. Both views are part of the BW Data Product Generator perspective, which grants quick access to both Eclipse views and gives users a singular entry point to all Data Product Generator functionalities in the BW Modeling Tools.

Prerequisites/Installation

As a prerequisite for using the Scenario Generator and the Data Subscriptions views, you need to install the BW Modeling Tools in the latest version or upgrade your existing installation to the latest version BWMT 1.27.

To use the BW Data Product Generator perspective, you need to at least upgrade the BW Modeling Tools to the patch version BWMT 1.27.1.

The new version can be downloaded from **SAP Marketplace** or SAP Toolspage.

After first-time installation you need to connect your BW system to the BW Modeling Tools interface by creating an associated BW project.

For more information on the installation and upgrade process, see Installation Guide for BW Modeling Tools.

Accessing the BW Data Product Generator Perspective

The BW Data Product Generator perspective provides BW modelers with a comprehensive interface to quickly access all Data Product Generator-related capabilities within the BW Modeling Tools, including the Eclipse views for the Scenario Generator and the Data Subscriptions.

The perspective opens with a welcoming screen showing further information about the Data Product Generator and instructions for first usage. In addition, it does open the two Eclipse views by default. However, the Eclipse views can also be opened and used independently from this on their own.

To access the BW Data Product Generator perspective, go to **Window** > **Perspective** > **Open Perspective** > **Other...** and select **BW Data Product Generator**.

Alternatively, you can also open it by selecting the **Open Perspective** icon ( ) in the top bar and choosing **BW Data Product Generator**.

Accessing the Eclipse Views

In addition to being part of the BW Data Product Generator perspective, the Eclipse views can also be opened and used on their own.

To access the Eclipse views on their own, go to **Window** > **Show View** > **Other...**:

- Select Business Warehouse > Data Subscriptions to open the Data Subscriptions view.
- Select Business Warehouse > Scenario Generator (Data Subscriptions) to open the Scenario Generator view.

Alternatively, you can also find the views via search, which can be opened by using CTRL+3 or by selecting the search icon ( ).

Scenario Generator View

With the Scenario Generator view you can select existing BW models and create scenarios out of them, as well as generate all required data subscriptions to represent these BW models in one operation.

The Scenario Generator view consists of three tabs:

- Scenario Core Objects: Select the BW objects you want to create a scenario for and make general settings for scenario creation. If you choose Collect, the Scenario Generator will collect all dependent objects for the selected top-level objects and create the scenario tree, visible in Object Relations.
- Object Relations:
  - Collected Objects: View the scenario tree for the selected BW objects after you have collected all dependent objects. Here, you can also view whether any of the top-level objects or dependent objects have existing data subscriptions.
  - Selection to Generate Data Subscriptions: Select data subscriptions to be generated to reconstruct the BW models, assign them to packages, and change their properties. If you choose Generate, the Scenario Generator will create all these data subscriptions in one operation.
- Messages: View all error messages, warnings, and confirmations related to operations performed by the Scenario Generator. This lets you debug in case you encounter errors while creating a scenario or generating the data subscriptions.

For more information on the workflow, see Using Scenario Generator.

Data Subscriptions View

With the Data Subscriptions view you can get an overview of existing data subscriptions in your BW system, as well as mass maintain them.

The main screen of the Data Subscriptions view lists all available data subscriptions for one BW project.

In the field BW Project you can choose for which BW project you want to display the data subscriptions. A BW project corresponds with a connection to a specific BW system.

Data Subscriptions List

The columns in the data subscriptions list display information about certain properties of the data subscriptions.

Columns in Data Subscriptions List

Column	Value
Subscription	Information about the data subscription. • Technical name of the data subscription, shown greyed out and in brackets before the description. • Description of the data subscription which has been defined by the customer.

Column	Value
	• InfoProvider associated with the data subscription, shown in braces behind the description.
Status	Shows whether the data subscription has been activated or not.
Package	Package to which the data subscription belongs to. The default package is the local package $TMP. which indicates the object is local and cannot be transported to other systems.
Master Language	Original language in which the data subscription was created. Descriptions, texts, and metadata are mainly in that language. • E - Object has been created in English. • D - Object has been created in German.
Scenario	Scenario to which the data subscription belongs to. **i Note** This column is only visible if you have selected Read Scenario References in Data Subscriptions Preferences.
Source Object	Name of the ODP-based source (InfoProvider) selected for the data subscription.
Target Name	Name of the target selected for the data subscription.
Delta Runtime	Shows whether delta extraction has been configured for data subscription. • **Full** - Full extraction has been set. • **Delta** - Delta extraction has been set.
Owner	User who owns the data subscription.
Last Changed By	User who last changed the data subscription.
Last Changed At	Date and time at which the data subscription has last been changed.
Last Run	Shows whether data subscription has already been run and the result of the last run. • No Run ( ) - Data subscription hasn't been run yet. • Success ( ) - Data subscription has been run yet and last run was successful. • Error ( ) - Data subscription has been run yet, but last run has failed. **i Note** This column is only visible if you have selected Show Last Run Status in Data Subscription Preferences.

Column	Value
Local Table (File)	`CTRL + Click` on **Show Local Table (File)** to open the local table (file) in the SAP Datasphere tenant.
Local Table State	Shows the status of the local table (file) that has been created or deployed for the data subscription. **ℹ Note** This column is only visible if you have selected Show Local Table Status in Data Subscription Preferences.

Configurations for Data Subscriptions List

The data subscriptions list can be configured and customized based on certain criteria.

- In the Filter Pattern field above the list, you can enter any filter value to filter the displayed list.
- If you click on any of the column headers, you can sort the list based on the values in this field in ascending or descending order. Clicking on the column header again will reverse the sorting order based on the previous configuration.
- In the top bar of the Data Subscriptions view, you can access further options:
  - Refresh ( ) - Refresh the list of data subscriptions to see latest changes or results of performed actions.
  - Open Link in Browser ( ) - Open the main screen of the Data Subscriptions app in the SAP BW/4HANA Cockpit. (For more information, see Data Subscriptions App in SAP BW/4HANA Cockpit).
  - Preferences ( ) - Open the Data Subscriptions Preferences menu to adjust columns and information displayed in the list.
- If you right-click on the data subscriptions in the list, you can make additional settings:
  - Select Show Scenario Subscriptions Only to only display data subscriptions in the list which belong to a scenario, that is, have been created via the Scenario Generator.

    ℹ Note
    This option is only available if you have selected Read Scenario References in Data Subscriptions Preferences.

  - Select Preferences to open the Data Subscriptions Preferences menu as an alternative way to using the icon in the top bar.

Data Subscription Preferences

Menu Option	Result
Show Local Table Status	Adds the column Local Table Status to the data subscriptions list where you can view the status of the local tables (file) that have been created for the data subscriptions. **⚠ Caution** Only use this option if you really require this information. Retrieving and displaying the status of the local tables (file) can take significant time and processing power to complete and you might have to wait for a while till you can see results.

Menu Option	Result
Show Last Run Status	Adds the column Last Run to the data subscriptions list where you can view whether the data subscription already has been run and if that last run has been successful or not.
Show Active Version	Displays data subscriptions in A (Active) version in the list. By default the list displays data subscriptions in M (Modified) version. This setting changes the default display. **ℹ Note** You can see which version is displayed in the line below the list. Object version 'M' corresponds with the default setting of only M versions being shown. Object version 'A' corresponds with the alternate setting of A versions being shown.
Read Scenario References	Retrieve information about the associated scenarios of the data subscriptions (if they have been created via the Scenario Generator). This setting adds the Scenario column to the data subscriptions list, as well as makes it possible to use the options Show Scenario Subscriptions Only, Show Scenario, and Delete Scenario.
Read Only Own	Display only data subscriptions which have either been created or last changed by the user. This information can also be viewed in the columns Owner and Last Changed By.

Actions for Data Subscriptions

If you right-click on any of data subscriptions in the list, you can access a menu to perform actions on one or multiple selected data subscriptions.

→ Tip

If you press CTRL and click on the data subscriptions, you can select multiple data subscriptions at once and perform an action on these. This lets you mass maintain data subscriptions in the Data Subscriptions view, for example, delete multiple data subscriptions at once.

Actions for Data Subscriptions

Menu Option	Action
Find Parent Scenarios	Display a list of parent data subscriptions belonging to the same scenario as the selected data subscription.
Copy Object Name	Copy the technical name of the selected data subscription.
Copy Cell	Copy the cell content of the selected cell, for example, the package name or the owner name.
Copy Link	Copy the URL of the Maintain Subscription screen of the data subscription in the SAP BW/4HANA Cockpit.
Open in Browser	Open the data subscription in the SAP BW/4HANA Cockpit in the Maintain Subscription screen, where you can view further details as well as edit some of the properties.

Menu Option	Action
Activate	Activate the selected data subscriptions to make them executable after they have been created or changed. This changes their status from Inactive to Active. → **Tip** Select multiple data subscriptions at once via `CTRL + Click` to mass activate data subscriptions.
Deactivate	Deactivate the selected data subscriptions to make them not executable. This changes their status from Active to Inactive.
Refresh	Refresh the entry of the data subscription to see latest changes or results of performed actions.
Delete	Delete the selected data subscriptions. This doesn't delete the local table (file) if it has already been deployed. → **Tip** Select multiple data subscriptions at once via `CTRL + Click` to mass delete data subscriptions.
Delete Scenario	Delete the associated scenario of the data subscription. You need to confirm the deletion before you can carry out the action. ℹ **Note** This option is only available if you have selected Read Scenario References in Data Subscriptions Preferences.

Maintaining Data Subscriptions

The data subscriptions can be maintained in the Data Subscriptions app in the SAP BW/4HANA Cockpit, in the RSDPS transaction in the SAP GUI, as well as in the Eclipse views in the BW Modeling Tools.

You can create data subscriptions, create data subscriptions from template, edit data subscriptions, activate and deactivate data subscriptions, and delete data subscriptions.

In the BW Modeling Tools it's possible to mass generate data subscriptions based on BW model, as well as mass delete, mass activate, and mass deactivate data subscriptions. In the SAP GUI you can mass delete data subscriptions based on target.

For detailed procedures on these actions, see:

- Creating Data Subscriptions
- Creating Data Subscriptions from Template
- Editing Data Subscriptions
- Transporting and Migrating Data Subscriptions
- Deleting Data Subscriptions
- Mass Deleting Data Subscriptions by Target

For reference documentation about all available actions and settings, see:

- Data Subscriptions App in SAP BW/4HANA Cockpit
- RSDPS Transaction in SAP GUI
- BW Data Product Generator Perspective in BW Modeling Tools (BWMT)

Note

On SAP BW 7.5 systems you need to use the RSPDS transaction in the SAP GUI instead of the Data Subscriptions app in the SAP BW/4HANA Cockpit to maintain the data subscriptions. The BWMT integration is available on SAP BW 7.5 and on SAP BW/4HANA systems.

Creating Data Subscriptions

To extract data from InfoProviders in your BW system into local tables (file) in your SAP Datasphere tenant, you need to create a data subscription.

Prerequisites

- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. During the configuration, the BW administrator has created a space to receive data pushed by the Data Product Generator (BW file space). (For more information, see Configuring the Data Product Generator.)
- The BW administrator has created one or multiple targets for the Data Product Generator that can be selected for the data subscription. (For more information, see Configuring Targets for the Data Product Generator).

Creating Data Subscriptions in the SAP BW/4HANA Cockpit

In the SAP BW/4HANA Cockpit you can create data subscriptions in the Data Subscriptions app.

1. In the SAP BW/4HANA Cockpit, go to the section **Data Product Generator for SAP Business Data Cloud** and open the app **Data Subscriptions**.
2. Select **Create** among the buttons in the top bar.
3. In the field **Source Name**, select the BW InfoProvider you want to extract data from. If you click on , a value help will open where you can search for available InfoProviders in your BW system. Search results will show their name and description.

   Note

   The data subscriptions use the ODP (Operational Data Provisioning) framework to extract data. Therefore, the selected source object must be an ODP-based source. You can select InfoObjects, DataStore objects (DSOs), CompositeProviders, MultiProviders, InfoCubes, or Queries as InfoProviders.

4. In the field **Target Name**, select the target for the data subscription. If you toggle , you can view all available targets for your BW system including their name and description. For more information, see Configuring Targets for the Data Product Generator.
5. Select **Save** to save the data subscription.
6. Select the **Package** the data subscription will belong to for transport requests. If you click on , a value help will open where you can search for available packages in your BW system. Search results will show their name and description.

> **Note**
> By default, the package $TMP is selected. This is a special package dedicated to creating local objects that won't be transportable. Only leave this package selected if there is no need to transport the created data subscription to quality or production systems, such as for tests or prototyping.

7. Select the **Transport Request** the data subscription will belong to.
   - **Choose from requests in which I am involved:** Here you can preview a list of requests you're actively involved in on your BW system and select one of them for transporting the data subscription. This means, you've either created the request, have added objects to the request, or have been explicitly added as a user to the request.
   - **Create new request**: Here you can create an entirely new request for your transport scenario.
   - **Enter a request number:** Here you can select amongst all requests on your BW system by entering the request number and choosing **Browse**.

   If the data subscription has been saved successfully, you will see a confirmation message appearing on the screen.

8. The data subscription will now be created. During the creation some of the properties are automatically generated or filled in.
   - The technical name of the data subscription (**Technical Name**) will be automatically generated and cannot be changed.
   - The description of the data subscription (**Description**) will be filled in based on the description of the BW InfoProvider, but still can be adjusted afterwards.
   - The timestamp when the data subscription was created (**Last Changed At**) and user who created it (**Last Changed By**) will be automatically filled in by the system and cannot be adjusted.

     These details can be viewed in the header section of the **Maintain Subscription** screen. Here, you can also see the current status of the data subscription (**Active Version**), the current status of the local table (file) (**Local Table State)** and whether the local table (file) already has been filled with data or not (**Has Data**). (For more information on possible versions/states of the data subscription, see Transporting and Migrating Data Subscriptions.)

9. (Optional) In the **Maintain Subscription** screen you can now make further settings for your data subscription.
   - Go to the **General** tab and enter a custom description for your data subscription in the **Description** field.
   - Go to the **Settings** tab and select the **Extraction Mode**. You can choose between **Full** and **Delta**. For more information, see Setting Extraction Mode for the Data Product Generator.
   - Go to the **Settings** tab and open **Selections** to limit the data set to be extracted via the data subscription. A value help will open, displaying all fields for which you can limit the data set. Enter conditions for the fields and confirm via **OK** to apply the them.
   - Go to the **Settings** tab and select the **Process Chain Variant** if you want to use a process chain for the extraction. If you click on , a value help will open, displaying all available process variants in your BW system including their name, the ID of the associated process chain, the process chain description, the trigger description and condition, and the period of executions. You can also search for variants.

     > **Note**
     > The process type needs to be **Load Process and Postprocessing - Data Subscription**.

   - Go to the **Projections** tab to define from which fields in the InfoProvider the data subscription should extract data. By default all fields/the entire data set will be extracted (**Full Projection**). If you deselect **Full Projection**, the table with the fields will become editable. You can now add fields to the projection by choosing them via the checkbox in the first column. Key fields always need to be selected and therefore will not be editable.

10. Select **Activate** to activate the data subscription and make it executable. This will change its status from **Not Executable** to **Executable**.

    You will see a confirmation message appearing on the screen informing you that an SAP Datasphere entity will be created and deployed asynchronously. The local table (file) in SAP Datasphere will now be created with the data subscription ID as its qualified name. If you go to the tab **Datasphere Logs** you can monitor these operations and confirm their success or failure.

11. Select **Run** to run the data subscription manually if you don't use a process chain.

    You will see a confirmation message appearing on the screen stating that the data subscription has been started. The data will now be extracted from the selected InfoProvider into the local table (file).

12. Go to **View Executions** to view a log of recent runs of all subscription.

    Locate your data subscription via the column **Subscription** and view whether the execution was successful or not (**Status**), how many records have been transferred from the InfoProvider to the local table (file) (**Transferred Records**), and how long the extraction operation took (**Duration**).

13. Select the button **View Local Table (File)** to open the local table (file) with the extracted data in the SAP Datasphere tenant.

    i Note

    The local table (file) is stored in a dedicated BW file space of type SAP HANA Cloud, data lake Files (HDLFS) and is not editable. For each BW system a separate space must be created during the configuration of the Data Product Generator. You cannot create any objects in this space except for task chains.

    All objects in the space are read-only, but data management tasks (such as deleting data) are still possible. In addition, you can share the local tables (file) in this space to other spaces where you can use them for modeling. The structure of the local table (file) cannot be changed as long as the data subscription exists. This ensures that extractions via process chain will continue to work.

    For more information, see Viewing Local Table (File) and Getting Data Updates.

Creating Data Subscriptions in the SAP GUI

In the SAP GUI you can create data subscriptions in the RSDPS transaction.

1. Open your BW system in the SAP GUI and call transaction RSDPS.
2. Select **Create** (Create) among the icons in the top bar.
3. In the field **Source Name**, select the BW InfoProvider you want to extract data from. If you click on , a value help will open, displaying all available InfoProviders in your BW system including their name, description and type (shown through a small icon). Select the row for the InfoProvider you want to choose and confirm via .

   i Note

   The field **Source Type** will be automatically set to ODP (BW InfoProvider) as this is the only available source type for the Data Product Generator.

4. In the field **Target Name**, select the target for the data subscription. If you click on , a value help will open, displaying all available targets for your BW system including the name, description, and whether they're a preferred target (marked through an X). Select the row for the target you want to choose and confirm via .

   i Note

The field **Target Type** will be automatically set to HDLFS (SAP HANA Cloud, data lake Files) as this is the only available target type for the Data Product Generator.

5. Select **Save** ( ) in the top bar to save the data subscription.
6. Select the **Package** the data subscription will belong to for transport requests and confirm via **Save** ( ). Alternatively, you can also create the data subscription as a local object by clicking on **Local Object** instead.

ℹ Note

If you create the data subscription as a local object, it will be assigned to the special package $TMP. This means it won't be transportable and only stays in the current development system.

7. The data subscription will now be created. During the creation some of the properties are automatically generated or filled in.
   - The technical name of the data subscription (**Subscription**) will be automatically generated and cannot be changed.
   - The description of the data subscription (**Description**) will be filled in based on the description of the BW InfoProvider, but still can be adjusted afterwards.
   - The timestamp when the data subscription was created (**Last Changed At**) and user who created it (**Created By**) will be automatically filled in by the system and cannot be adjusted.
   - The owner of the data subscription (**Owner**) will be automatically filled in based on who created it and cannot be adjusted.
   - Further information about the BW InfoProvider will be automatically provided in the section **Source Object**. Here, you can now view the name of the InfoProvider (**Name**), the type of the InfoProvider (e.g. aDSO) (**Object Type**), the user who last changed the InfoProvider (**Last Changed By**) and the InfoArea it belongs to (**InfoArea**). None of these settings can be changed since they have been pulled from the metadata of the InfoProvider.
   - Further information about the local table (file) that will be created and deployed are now also available in the section **Target**. Here, you can view the technical name of the local table (file) (**Technical Name**), the inbound buffer path in the object store (**File Path**), as well as its CSV properties such as how meta information are saved (**Meta Information**) and what separator (**Separator**) and escape character (**Escape Character**) are used.
8. (Optional) In edit mode you can now make further settings for your data subscription.
   - Set the extraction mode between full extraction (**Full**) or delta extraction (**Delta**) in the field **Extraction Mode**. For more information, see Setting Extraction Mode for the Data Product Generator.
   - Toggle the scheduling settings between manual/direct execution of the data subscription (**No Scheduling**) and scheduling data subscription runs via process chain (**Process Chain**) in the field **Scheduling**.

     If you want to run the data subscription via process chain, you also need to enter the process variant in a pop-up field **Process Variant**. If you click on , a value help will open, displaying all available process variants in your BW system including their name, description, the name of the associated process chain, and the conditions and period for the execution. Select the row for the process variant you want to choose and confirm via .

     ### ℹ Note

     The process type needs to be **Load Process and Postprocessing - Data Subscription**.

   - Click on **Edit Selection** to limit the data set to be extracted via the data subscription. A value help will open, displaying all fields for which you can limit the data set. Enter conditions for the fields and confirm via to apply the filters.

- Click on **Edit Projection** to define from which fields in the InfoProvider the data subscription should extract data. A value help will open, displaying all available fields in the selected InfoProvider including their name, whether they're a key field, their description, their data type, and some further information. All selected fields are marked with an . If you click on the icon, you can add or remove the fields as needed. Confirm via to save the changes.

  ℹ Note

  Key fields always need to be selected in the projection.

9. Select **Save** ( ) in the top bar to save the changes.
10. Select **Activate Data Subscription** ( ) to activate the data subscription and make it executable. This will change its status from **Inactive** to **Active**.
11. Go back to the main screen, select the data subscription you have just created in the list, and choose **Run** (Run) in the top bar. This will start the data extraction according to the defined settings manually if you don't use a process chain.

    As a result, the local table (file) will be filled with the data from the BW InfoProvider. You can access the local table (file) as well as perform further actions on your data in your SAP Datasphere tenant. For more information, see Viewing Local Table (File) and Getting Data Updates.

Creating Data Subscriptions from Template

You can create a data subscription based on a template from an existing data subscription. This allows you to take over settings, enabling you to create data subscriptions faster and replicate existing extraction operations to more InfoProviders.

Creating Data Subscriptions from Template in the SAP BW/4HANA Cockpit

In the SAP BW/4HANA Cockpit you can create data subscriptions in the Data Subscriptions app.

1. In the SAP BW/4HANA Cockpit, go to the section **Data Product Generator for SAP Business Data Cloud** and open the app **Data Subscriptions**.
2. Select one of the data subscriptions in the list. This will be the data subscription you use as a template to create another data subscription.
3. Select **Create from Template** amongst the buttons in the top bar.

   The details and settings of the new data subscription will be filled in based on the information in the selected data subscription.

   ℹ Note

   The new data subscription will be a 1:1 copy of the old data subscription. However, it is possible to adjust a select amount of properties in the copied data subscription.

4. (Optional) In the field **Source Name**, select a different BW InfoProvider to extract data from with the same settings.
5. (Optional) In the field **Target Name**, select a different target to extract data into with the same settings.
6. Select **Save** to save the new data subscription.
7. Select the **Package** the data subscription will belong to for transport requests.

   ℹ Note

Even if you create a new data subscription based on a template from a data subscription of type SUBS, the new data subscription still will be of type DSUB and therefore will be automatically transportable.

8. (Optional) After you have saved the data subscription, you can adjust further settings you don't want to take over from the template. It's possible to change the description (Description), the extraction mode (Settings > Extraction Mode), the selection (Settings > Selections), the process chain (Settings > Process Chain Variant), and the projection (Projections).

9. (Optional) Select Save to save the data subscription with the adjusted properties.

Creating Data Subscriptions from Template in the SAP GUI

In the SAP GUI you can create data subscriptions in the RSDPS transaction.

1. Open your BW system in the SAP GUI and call transaction RSDPS.

2. Select one of the data subscriptions in the list. This will be the data subscription you use as a template to create another data subscription.

3. Select Create from Template (Create from Template) amongst the icons in the top bar.

   The details and settings of the new data subscription will be filled in based on the information in the selected data subscription.

   Note

   The new data subscription will be a 1:1 copy of the old data subscription. However, it is possible to adjust a select amount of properties in the copied data subscription.

4. (Optional) Adjust properties of the new data subscription you don't want to take over from the template. It's possible to change the selected BW InfoProvider (Source Name), the selected target (Target Name), the extraction mode (Extraction Mode), and the scheduling settings (Scheduling).

5. Select Save ( ) in the top bar to save the new data subscription based on the template.

6. Select the Package the data subscription will belong to for transport requests.

   Note

   Even if you create a new data subscription based on a template from a data subscription of type SUBS, the new data subscription still will be of type DSUB and therefore automatically transportable.

7. (Optional) After you have saved the data subscription, you can adjust further settings. It's now possible to change the description (Description), the selection (Edit Selection), and the projection (Edit Projection). In addition, you can now also still adjust the extraction mode (Extraction Mode) and scheduling settings (Scheduling).

8. (Optional) Select Save ( ) in the top bar to save the data subscription with the adjusted properties.

Editing Data Subscriptions

You can edit existing data subscriptions after you have created them. This allows you to still adjust the settings of a data subscription after initial creation.

Editing Data Subscriptions in the SAP BW/4HANA Cockpit

In the SAP BW/4HANA Cockpit you can edit data subscriptions in the Data Subscriptions app.

1. In the SAP BW/4HANA Cockpit, go to the section **Data Product Generator for SAP Business Data Cloud** and open the app **Data Subscriptions**.

2. Click on the data subscription you want to edit in the list.

   This opens the Maintain Subscription screen where you can view read-only properties of the data subscription, as well as adjust the editable ones. In contrary to the SAP GUI, here you are automatically in edit mode and don't need to switch to a dedicated edit screen to perform changes.

3. In the Maintain Subscription screen, adjust properties and settings as needed.

   i Note

   After you have created the data subscription and not run yet, you can still edit the description (Description), the process chain (Settings > Process Chain Variant), the extraction mode (Settings > Extraction Mode), the projection (Projections), and the selection (Settings > Selections).

   Once you have run the data subscription and the local table (file) has been created and deployed, you can only edit the description (Description), and all other fields (such as extraction mode, process chain, ...) become read-only and cannot be adjusted anymore.

4. Select Save to save the changes.

5. Select Activate to make the data subscription executable again.

   i Note

   In the Maintain Subscription screen you are viewing and editing the M (Modified) version of the data subscription. If you do any changes, you have to activate the data subscription again, so that the A (Active) version equals the M (Modified) version. For more information on the versions, see Transporting and Migrating Data Subscriptions.

Editing Data Subscriptions in the SAP GUI

In the SAP GUI you can edit data subscriptions in the RSDPS transaction.

1. Open your BW system in the SAP GUI and call transaction RSDPS.

2. Select the data subscription you want to edit from the list.

3. Choose **Edit** (Edit) among the icons in the top bar.

   As a result, the edit screen will open, providing an overview of the fields you can edit for the selected data subscription.

4. In the edit screen, adjust properties and settings as needed.

   i Note

   Similarly to the Data Subscriptions app, you can still edit the description (Description), the scheduling settings (Scheduling), the extraction mode (Extraction Mode), the projection (Edit Projection), and the selection (Edit Selection) if you have not run the data subscription yet.

   Once you have run the data subscription and the local table (file) has been created and deployed, you can only edit the description (Description). All other fields (such as extraction mode, scheduling settings, ...) have now become read-only and cannot be adjusted anymore.

   → Tip

In the edit screen you can also access the following options in the top bar:

- **Display ↔ Change** ( ): Switch between displaying the properties of the data subscription in view-only mode and in edit mode. This allows you to see directly which properties you can still change (fields displayed in white) and which not (fields greyed out).
- **Check Data Subscription** ( ): Check the data subscription for any potential issues which could impact the extraction of data. The results can be viewed in a log screen, including any warnings or error messages concerning the status of the source (InfoProvider), the target, the connection to SAP Datasphere, and the local table (file).
- **Activate Data Subscription** ( ): Make the data subscription executable after it has been created or changed. This changes the status of the data subscription from **Inactive** to **Active**.
- **Data Preview** (Data Preview): Preview the data set associated with the data subscription. It's also possible to download this data set into a local file or export it into other formats.

5. Select **Save** ( ) in the top bar to save the changes.
6. Select **Activate Data Subscription** ( ) to make the data subscription executable again.

ℹ Note

In the edit screen you are viewing and editing the M (Modified) version of the data subscription. If you do any changes, you have to activate the data subscription again, so that the A (Active) version equals the M (Modified) version. For more information on the versions, see Transporting and Migrating Data Subscriptions.

Transporting and Migrating Data Subscriptions

You can transport data subscriptions between different BW systems (e.g. production, development, quality) if they are in the DSUB format (TLOGO type for the data subscriptions). To migrate the data subscriptions from the SUBS to the DSUB format, you can use a dedicated ABAP report in the SAP GUI.

Transporting Data Subscriptions

To make the data subscriptions transportable, a new data subscriptions type (DSUB) has been introduced. The DSUBs follow the TLOGO framework for all BW objects with DSUB denoting the TLOGO type.

ℹ Note

The TLOGO framework (TLOGO = Transportable Logical Object) is the unified transport architecture behind all BW objects, letting you group, identify, and transport BW objects through the system landscape.

Transportable BW objects have a TLOGO type, such as ELEM for BW Queries, or DSUB for the data subscriptions. The TLOGO type defines what kind of object you want to transport, how it is stored in the database, how it behaves during the transport, and which sub-objects belong to it. This allows the BW system to treat complex BW objects as single units during transport.

Local data subscriptions (which are not transportable) remain in the old data subscriptions type (SUBS). You can migrate data subscriptions from SUBS to DSUB format by using the ABAP report RSDDS_DSUB_MIGRATION. In addition, all new data subscriptions are now automatically created as DSUBs. For more information on the data subscriptions types, see Data Subscription (DSUB and SUBS).

To be able to transport a data subscription, it needs to fulfill the following prerequisites:

- Be of the type DSUB (TLOGO type) and not SUBS. If you want to transport a data subscription of type SUBS, you first need to migrate it to the DSUB type via RSDDS_DSUB_MIGRATION. (For more information, see section "Migrating Data Subscriptions".)
- Be assigned to a transport request and package. This happens during initial creation of the data subscription (Creating Data Subscriptions) or during migration to DSUB format.

  ℹ Note

  A transport request is a container in the Change and Transport System (CTS) that collects the changes in the BW object you want to move between different systems.

  The package is the ABAP container the BW object is assigned to in the repository. It determines whether the object is transportable, as well as its transport layer. The package $TMP is for holding local objects which cannot be transported. It is the default package assigned if you create a new data subscription. If you want to transport the data subscription, ensure to assign it to a different package than $TMP.

  If no package is given for the data subscription, it will be automatically assigned to $TMP.

- Be in A (Active) state, which means the data subscription must have been activated. (For more information, see section "Activating and Deactivating Data Subscriptions".)

Activating and Deactivating Data Subscriptions

The TLOGO object for the data subscriptions (DSUB) supports an A (Active) and an M (Modified) version:

- **A (Active)**: This version of the data subscription is validated and executable. All changes have been checked in and runtime artifacts have been generated or regenerated. Data subscriptions in A state can be run or transported to production and quality systems.
- **M (Modified)**: This version of the data subscription has been changed and is still marked as a work in progress. Before you can run or transport it, you need to activate it to convert it to A state and check in the changes.

Activating the data subscription turns it from an M (Modified) to an A (Active) state. Deactivating the data subscription sets it in a state that cannot be run.

There are different ways to activate or deactivate a data subscription:

- **Data Subscriptions app (SAP BW/4HANA Cockpit):**
  - Select a data subscription in the list and choose **Toggle Status** in the top bar to switch between active and inactive version.
  - Click on a data subscription to open the **Maintain Subscription** screen and there choose **Activate** or **Deactivate** in the bottom bar.

    ℹ Note

    In the field **Active Version** in the data subscription details header, you can view the current state of the object:

    - **Executable, Equal to Revised Version:** Data subscription has been activated and A (Active) version equals M (Modified) version. It can be run or transported.
    - **Executable, Not Equal to Revised Version:** Data subscription has been activated, but changed afterwards, so A (Active) version is different than M (Modified) version. It can be run in the A version, but if you want to consider changes, you first need to activate again.

- **Not Executable, Not Equal to Revised Version:** Data subscription has been deactivated and changed afterwards, so A (Active) version is different than M (Modified) version.
- **Not Executable, Equal to Revised Version:** Data subscription has been deactivated, but no changes have been made yet, so A (Active) version equals M (Modified) version.
- **Does not Exist:** Data subscription only exists in M (Modified) version and has never been activated before. This is typically the case after first time creation of the object.

- **RSDPS transaction (SAP GUI):**
  - Select a data subscription in the list and select **Edit** (Edit) to open the edit screen. Here choose **Activate Data Subscription** ( ) to activate the data subscription.

    ℹ Note

    In the field **Active Version** in the edit screen, you can view the current state of the object:

    - **Not exists:** Data subscription has never been activated before. This is typically the case after first time creation of the object.
    - **Executable:** Data subscription has been activated and can be run or transported.
    - **Not executable:** Data subscription has been deactivated and cannot be run or transported.

    The status of the data subscription can also be viewed in the columns **State** and **Revision** in the data subscriptions list. For a full overview, see RSDPS Transaction in SAP GUI.

Migrating Data Subscriptions from SUBS to DSUB

You can use the RSDDS_DSUB_MIGRATION report in the SAP GUI to convert existing local data subscriptions (SUBS) into transportable data subscriptions (DSUB).

ℹ Note

All newly created data subscriptions are automatically DSUBs. You can only transport data subscriptions if they are DSUBs and not SUBSs, so performing the migration is a necessary prerequisite for the transporting operation.

To migrate the data subscriptions, perform the following procedure:

1. Call transaction SE38 and enter the program name RSDDS_DSUB_MIGRATION, then select **Execute** ( ) in the top bar.

   As a result, the report **TLOGO Migration for Data Subscriptions** will open.

2. Select the data subscriptions you want to migrate from SUBS to DSUB format. You can select one data subscription or multiple at once.

3. In the field **Transport Request**, enter the number of the transport request for the DSUBs.

   ℹ Note

   If you do not already have created a transport request, you can do so via SE09/SE10. Make sure to assign the correct package to the request so the transport layer knows the system path.

4. In the field **Package**, choose the package the objects will belong to. If you click on , a pop-up will open in which you can specify the package, as well as its transport layer (system path for objects in the package), and the maximum number of hits (how many objects can be assigned to the package).

5. (Optional) Select **Enable Simulation** if you want to preview the simulated results of the migration operation without performing any actual changes on your objects.

As a result, all selected data subscriptions in the SUBS format will be migrated to be TLOGOs in the DSUB format.

They can now be transported between different systems, as well as activated to switch between M (Modified) and A (Active) version.

Deleting Data Subscriptions

You can delete existing data subscriptions in the Data Subscriptions app or in the RSDPS transaction. This allows you to remove unused or outdated data subscriptions from your systems.

Deleting Data Subscriptions in the SAP BW/4HANA Cockpit

In the SAP BW/4HANA Cockpit you can delete data subscriptions in the Data Subscriptions app.

1. Open the SAP BW/4HANA Cockpit.
2. Go to the section **Data Product Generator for SAP Business Data Cloud** and open the app **Data Subscriptions**.
3. Select the data subscription you want to delete from the list.
4. Choose **Delete** among the buttons in the top bar.
5. (Optional) If you have already run the data subscription and a local table (file) has been created and deployed in SAP Datasphere, log on to SAP Datasphere and delete the local table (file) as well.

   ℹ Note

   You can only select and delete one data subscription at once. It is not possible to delete multiple data subscriptions in one operation.

Deleting Data Subscriptions in the SAP GUI

In the SAP GUI you can delete data subscriptions in the RSDPS transaction.

1. Open your BW system in the SAP GUI and call transaction RSDPS.
2. Select the data subscription you want to delete from the list.
3. Choose **Delete** (Delete) among the icons in the top bar.
4. (Optional) If you have already run the data subscription and a local table (file) has been created and deployed in SAP Datasphere, log on to SAP Datasphere and delete the local table (file) as well.

   ℹ Note

   Similarly to the Data Subscriptions app, it is not possible to delete multiple data subscriptions at once. You can only do this by using the ABAP report RSDDS_DELETE_BY_SOURCE (Mass Deletion for Data Subscriptions by Target) which deletes all data subscriptions belonging to one target. The program is only accessible in the SAP GUI. For more information, see Mass Deleting Data Subscriptions by Target.

Deleting Dependent Objects

Deleting the data subscription does not automatically delete the corresponding local table (file) as well. If the data subscription has been deleted, the local table (file) will remain in the SAP Datasphere tenant. However, there it will now become editable (losing the read-only tag) and can be deleted manually.

> → **Tip**
>
> In case of an offloading scenario it is not necessary to delete the local tables (file). Here, you move your BW data to SAP Business Data Cloud in a one-time operation and delete it afterwards in your BW system.

The data products created out of the BW data in the local tables (file) do not need to be deleted since they are released once and then need to be kept stable.

Mass Deleting Data Subscriptions by Target

You can mass delete data subscriptions if they belong to the same target (BW file space in SAP Datasphere or SAP Datasphere tenant). This allows for easier maintenance and to clear up your BW system of data subscriptions if you don't need them anymore.

Accessing Deletion Report

The mass deletion can be performed with a dedicated ABAP report in the SAP GUI.

You can access the report in two different ways:

1. Call transaction SE38 and enter the program name RSDDS_DELETE_BY_SOURCE, then select **Execute** ( ) in the top bar.
2. Directly access the report in the RSDPS transaction by selecting **Goto** > **Delete Data Subscriptions by Target**

As a result, the report **Mass Deletion of Data Subscriptions by Target** will open.

Deleting Data Subscriptions

To mass delete the data subscriptions based on target, perform the following procedure:

1. Go to **Selection Parameters** and select the **Target**.

   You can do so either by entering the name of the target into the field or by selecting to open the value help.

   In the value help you can see a list of all targets configured for your BW system including their names and descriptions. If you have configured a user-preferred target, it will be marked up with an X in the column **Preferred Target**. Here, select the row with the target you want to choose and then confirm via .

2. (Optional) Go to **Deletion Options** and select **Simulate Deletion** if you want to test the deletion operation based on the selected target without actually deleting any data subscriptions in your production system.
3. (Optional) Go to **Deletion Options** and select **Force Deletion** if you want to force delete the data subscriptions while ignoring all errors and validations during the deletion operation (with the exception of transport/authority/lock scenarios).

   > → **Tip**
   >
   > A common use case for force deletion is if you want to delete the data subscriptions even if the associated SAP Datasphere tenant or BW file space doesn't exist anymore. This is, for example, the case if you had a SAP Business Data

Cloud trial with a limited lifetime and want to remove the corresponding data subscriptions from your BW system afterwards.

⚠ Caution

Before you carry out a force deletion, check your connection to SAP Datasphere. If the connection is not working during the force deletion, the local tables (file) might not lose their read-only tag, which means they cannot be deleted afterwards in SAP Datasphere.

4. Select **Execute** ( ) in the top bar of the report to perform the deletion operation.

   Confirm that you want to delete all data subscriptions for the selected target while understanding that the deletion operation cannot be undone.

As a result, the data subscriptions will be mass deleted according to the selections you have made.

You can view the outcome of the deletion operation in a log screen that appears after the deletion finished. Here you can also see how many data subscriptions have been deleted and any error messages or issues that might have come up.

If you have selected **Simulate Deletion** the log screen will show a preview of the results of the deletion operation that would be carried out.

Using Scenario Generator

The Scenario Generator is an Eclipse view in the BW Modeling Tools (BWMT) that is part of the BW Data Product Generator perspective. It lets you reconstruct existing BW models as local tables (file) in SAP Datasphere by generating the required data subscriptions through the creation of a scenario.

Important Concepts

A **scenario** is a list of references between data subscriptions which are required to reconstruct a specific BW model (such as a DataStore object (advanced) (aDSO) or a CompositeProvider (HCPR)). For example, if you have a CompositeProvider, you would not only need a data subscription to represent the top-level object, but also further data subscriptions for the InfoObjects it contains and other dependencies.

The **scenario tree** saves these dependencies between the objects in the BW model, as well as the equivalent data subscriptions and their dependencies. It also considers existing data subscriptions for dependent objects to avoid the creation of duplicate data subscriptions and local tables (file).

The **Scenario Generator** lets you not only create scenarios out of your BW models, but also mass generate the associated data subscriptions out of the scenario tree. Through this you can replicate an entire BW model into local tables (file) in your SAP Datasphere tenant in SAP Business Data Cloud. There, the extracted data and associated semantics are stored in table format and become available for further use cases, such as creating data products or sharing to SAP Databricks.

Key Workflow

There are two steps needed to convert the BW model into equivalent local tables (file):

1. Creating Scenarios: Create a scenario by selecting any existing BW objects from your system and collecting its dependent objects and existing data subscriptions.

2. Generating Data Subscriptions: Use the scenario to generate all required data subscriptions for the BW object and its dependency tree. Afterwards, the data subscriptions need to be run to extract the data and semantics into local tables

(file).

For a detailed reference of the Eclipse view, see BW Data Product Generator Perspective in BW Modeling Tools (BWMT).

Creating Scenarios

You can create a scenario for a BW model in the Scenario Core Objects tab of the Scenario Generator view in the BW Modeling Tools. By creating a scenario, all dependent objects and existing data subscriptions for the BW model are collected and the scenario tree is built.

Prerequisites

- The BW modeler user has installed the BW Modeling Tools in the latest version or upgraded the existing installation to the latest version and can access the Scenario Generator and Data Subscriptions views in the BW Modeling Tools. The minimum required version is BWMT 1.27. (For more information, see BW Data Product Generator Perspective in BW Modeling Tools (BWMT).)
- The BW modeler user has created BW objects, such as InfoObjects, DataStore objects (advanced) (aDSOs), or CompositeProviders which are available in the BW Modeling Tools.
- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. (For more information, see Configuring the Data Product Generator).

Context

In the **Scenario Core Objects** tab of the Scenario Generator you can create new scenarios by adding BW objects (such as InfoObjects, DataStore objects (advanced) (aDSOs) and CompositeProviders) from your BW system to the object list. Via **Collect** you can built the scenario tree out of the BW objects in the object list. Depending on the settings you have made, existing data subscriptions and dependent objects of the BW models are collected and shown as dependencies of the top-level objects.

On a scenario-level you can also make general settings for the data subscriptions to be generated for the scenario, including choosing the target and the process chain variant. Further settings for the data subscriptions can be made once you have built the scenario tree and the data subscriptions list.

Procedure

1. Open the BW Modeling Tools and access the Scenario Generator view, either by searching for it via CTRL+3 or selecting it under **Window** > **Show View** > **Other...** > **Business Warehouse** > **Scenario Generator (Data Subscriptions)**.
2. Go to the tab **Scenario Core Objects**.
3. Select **Add BW Object** ( ) by clicking on the icon in the top bar or by right-clicking on the objects list and choosing the menu option.
4. Search for the BW objects you want to add and confirm your selection. You can add one or multiple BW models to the scenario of all previously existing ones in your BW system.

   Added objects will appear in the object list on the left side. List entries include the name and the type of the BW object.

   If you right-click on the object list, you can access further options:

   - **Add Object List**: Open a dialogue to access all saved object lists on your BW system, download these lists as files to your local machine to save them permanently in-between sessions, and restore lists previously saved as files.
   - **Copy List**: Copy the currently displayed object list to the clipboard. If you copy the current list, you can also access it in the **Add Object List** screen where you can save it permanently on your local machine to restore it in later sessions in the BW Modeling Tools.

- **Paste List**: Paste the previously copied list from the clipboard.
- **Remove**: Remove the selected BW object from the list again.

5. Once you have completed the object list with all BW objects you want to include in the scenario, you can make further settings to build the scenario tree:

   a. Select the **Collection Grouping** to define whether the scenario tree should only be built from the selected BW objects or should also include further objects along the data flow before or afterwards.

Collection Grouping	Result
Only Necessary Objects	The scenario will only include the necessary objects in the context of the selected BW objects. This is the recommended mode.
Data Flow Before and Afterwards	The scenario tree include the selected BW objects and any other objects related to them upstream or downstream in the data flow.
Data Flow Before	The scenario will include the selected BW objects and any other objects related to them upstream in the data flow.
Data Flow Before (Direct)	The scenario will include the selected BW objects and any other objects directly related to them upstream in the data flow.
Data Flow Afterwards	The scenario will include the selected BW objects and any other objects related to them downstream in the data flow.

   b. Select the **Association Levels** to define the amount of levels included in the dependency tree. The default number of levels is 4. The minimum number is 1 and you can adjust the setting counting upwards in steps of 1.

6. Select the target for the data subscriptions in the scenario. Via **Browse** you can choose from all existing targets in your BW system.

ℹ Note

You can still adjust the target for the individual data subscriptions later on in the **Selection to Generate Data Subscriptions** section in the **Object Relations** tab. (For more information, see Generating Data Subscriptions.)

7. (Optional) Select the process chain variant for the scenario. Via **Browse** you can choose from all available variants for running the data subscriptions.

ℹ Note

While it's mandatory to select a target, you don't need to select a process chain variant. If you leave the field empty, the data subscriptions can be run manually and are not executable via process chain.

8. (Optional) Tick off the check boxes for further settings for the scenario:

Setting	Result
Perform Target Check	Checks connection for selected target in advance. This allows to avoid or resolve connection errors due to faulty target configuration before even trying to generate new data subscriptions for the scenario. You are forced to select a valid or correctly connected target, making operations with the scenario easier and less prone to throwing errors.

Setting	Result
Consider Transport Configuration	Activates transport handling for data subscriptions in the scenario. This allows to assign packages to the data subscriptions other than $TMP (local package), which makes them transportable between systems.
Use Existing Data Subscriptions	Includes existing data subscriptions in the scenario tree so they won't be duplicated. For example, if you want to rebuilt a DataStore object (advanced) (aDSO) which includes the InfoObject 0COUNTRY and the InfoObject already has an existing data subscription in the BW system, it can be reused for the scenario and doesn't need to be newly generated. If you select this option the scenario tree in **Collected Objects** will also show the existing data subscriptions for the objects.
Hide Non-Relevant Objects	Hides objects from the scenario tree for which you can't create any data subscriptions, such as 0CALDAY. This allows to build a cleaner scenario tree. While all dependent objects are always collected, they don't necessarily need to be shown in the scenario tree, which is why this option is active by default.

9. Choose **Collect** to collect all top-level objects and dependent objects of the selected BW models and create the scenario.
10. Switch to the tab **Object Relations** and go to **Collected Objects** to view the scenario tree.

    You can toggle dependent objects and existing data subscriptions for the top-level objects. Existing data subscriptions are only shown in the table if you have selected **Use Existing Data Subscriptions** in the scenario options. The table shows more details about the collected objects and data subscriptions in several columns.

Column	Value
Object Name	Technical name of the BW object, dependent object, or data subscription.
Data Subscription	If a data subscription is already existing for the object, this column will receive a green check mark. **Note** The green check marks will only be visible, if you have selected **Use Existing Data Subscriptions** in the scenario options.
Target Name	The target of the data subscription, if it already exists for the object. **Note** This column will only be used if you have selected **Use Existing Data Subscriptions** in the scenario options.
Process Chain Variant	The process chain variant of the data subscription, if it already exists for the object. **Note** This column will only be used if you have selected **Use Existing Data Subscriptions** in the scenario options.

Column	Value
Association Type	How the top-level object and dependent object are related to each other.

Results

You have created the scenario with the selected BW objects and settings. If you want to generate the corresponding data subscriptions, you need to select **Generate** in the **Object Relations** tab. (For more information, see Generating Data Subscriptions.)

Generating Data Subscriptions

You can generate the data subscriptions for the scenario in the Object Relations tab of the Scenario Generator view in the BW Modeling Tools. Through this you can create all data subscriptions of the scenario in at once in a single operation.

Prerequisites

- The BW modeler user has installed the BW Modeling Tools in the latest version or upgraded the existing installation to the latest version and can access the Scenario Generator and Data Subscriptions views in the BW Modeling Tools. The minimum required version is BWMT 1.27. (For more information, see BW Data Product Generator Perspective in BW Modeling Tools (BWMT).)
- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. (For more information, see Configuring the Data Product Generator).
- The BW modeler has created a scenario out of one or multiple BW models and collected all dependent objects and references. (For more information, see Creating Scenarios).

Context

In the **Object Relations** tab of the Scenario Generator you can generate all required data subscriptions for a scenario. In addition to the general settings you have made on scenario-level, you can customize settings for the individual data subscriptions including description, extraction mode, process chain variant, selections, projection, target, and package.

All data subscriptions for the scenario are shown in a list under **Selection to Generate Data Subscriptions**. Here, you can view a select amount of their properties as well as perform mass operations, such as mass assigning new packages. The button **Generate** starts the operation to generate the data subscriptions for the scenario. The results of the operation will be shown in the **Messages** tab of the Scenario Generator.

Procedure

1. Open the BW Modeling Tools and access the Scenario Generator view, either by searching for it via CTRL+3 or selecting it under **Window** > **Show View** > **Other...** > **Business Warehouse** > **Scenario Generator (Data Subscriptions)**.
2. Go to the section **Selection to Generate Data Subscriptions** in the tab **Object Relations**.

   Here, you can preview all data subscriptions that will be created for the selected BW models. The list doesn't only show data subscriptions for the top-level objects, but also for dependent objects, which you can see if you toggle the data subscription of the top-level object.

   The data subscriptions list has the following columns with more details:

Column	Information
Data Subscription	Shows the preliminary technical name of the data subscriptions in brackets, as well as the description, and the name of the source object in braces. The preliminary technical name is auto-generated and cannot be changed. The description is customer-defined and can still be edited.
Status	Shows the status of the to-be-created data subscriptions. indicates the data subscription is new and will be freshly created if you perform the operation.
Selections	Shows if you have defined selections for the data subscription. With a selection you can limit the dataset to be extracted from the InfoProvider by selecting certain fields and adding conditions for extracting data from them. indicates selections have been defined, otherwise the field will be empty.
Projection	Shows if you have defined a projection for the data subscription. With a projection you can limit the dataset to be provisioned from the InfoProvider by selecting certain fields that should be included or excluded. indicates a projection has been defined, in case of a full projection (default setting) the field will be empty.
BW Object	Shows the name of the source object (BW InfoProvider) for the data subscription, as well as the object type. The object type is visible as an icon in front of the object name and as the TLOGO type in braces after it (e.g. ADSO, IOBJ). The BW object cannot be edited anymore, as it is set via the scenario tree.
Target Name	Shows the name of the target you have selected for the data subscription. The target shown here will be the one you have selected generally for the scenario. You can still edit the targets as long as you ensure all the data subscriptions in the scenario have the same target.
Package	Shows the name of the package the data subscription belongs to. The $TMP package is the local package and indicates the data subscriptions are local objects and cannot be transported. To make them transportable, you need to select a different package. Packages can be assigned individually or mass assigned to multiple data subscriptions at once in the list.

Under the list you can see the overall number of data subscriptions in the scenario, as well as the number of references between them or other existing data subscriptions.

Via the field **Filter Pattern** you can filter the list according to the entered filter value.

3. Select the **Mode** for the data subscription generation.

Mode	Result
No Activation	The data subscriptions won't be activated during the operation and you need to activate them manually afterwards.
Activate Directly	The data subscriptions will be activated directly during the operation and are immediately executable.
Activate in Background	The data subscriptions will be activated in the background during or after the operation and will be executable afterwards.

⚠ Caution

The option **Activate Directly** takes a lot of time and processing power, so choosing **Activate in Background** is nearly always preferred to save computing resources.

4. (Optional) Edit individual data subscriptions in the data subscriptions list.

Before generating the data subscriptions, you can still edit the settings for individual data subscriptions in addition to having made generalized settings in the **Scenario Core Objects** tab.

Select a data subscription and right-click to access further options:

- **Edit**: Change advanced settings of the data subscription.

Setting	Options
Description	Change the description of the data subscription.
Package	Select a new package for the data subscription. Via **Browse** you can search for existing packages in your BW system.
Extraction Mode	Choose between **Full** and **Delta** as extraction mode. For more information, see Setting Extraction Mode for the Data Product Generator.
Process Chain Variant	Select a process chain if you want to schedule the extraction operation of the data subscription. Via **Browse** you can select from all existing process chain variants in your BW system. Via **Remove** you can remove the process chain again to run the data subscription manually.
Target Name	Select a different target for the data subscription. Via **Browse** you can choose from all existing targets in your BW system. ⚠ **Caution** The data subscriptions belonging to one scenario must all have the same target, otherwise you will receive an

Setting	Options
	error message if you try to generate the data subscriptions and cannot complete the operation.
Selections	Click on **Edit** to define a new selection for the data subscription. On the left side, select the field for which you want to limit the to-be-extracted dataset. Under **Add New Selection**, enter the limiting condition and value: ▪ **Equal**: Enter a certain value the field value must be equal to. ▪ **Between**: Enter a certain value range the field value must lie in between. ▪ **Starts With**: Enter a certain value the field value must start with. Select **Add** to add the new selection. You can add multiple selections. The selections including the field, limiting condition, and value will be shown in a list on the right side. If you right-click on the list, you can remove a specific selection again via **Remove**, or remove all selections at once via **Remove All**. Once you're finished, select **Apply** to apply the defined selections.
Projections	Click on **Edit** to add a projection for the data subscription. The default setting is **Full Projection**, which means the full dataset will be provisioned. To define a different projection, deselect **Full Projection**. You can now select the provisioned dataset from all existing fields. The key field is marked up with a key icon and must be selected as a mandatory field. Once you're finished, select **Apply** to apply the defined projection.

- **Copy Name**: Copy the technical name of the data subscription to the clipboard.
- **Replace**: Replace the data subscription with another one matching the selected criteria. The new data subscription must have the same BW project, source object, and target as the old one.
- **Replace Multiple**: Replace multiple data subscriptions in the list at once with another one matching the selected criteria. The new data subscription must have the same BW project, source object, and target as the old ones. In the selection dialog you can also preview details such as projection and selection to match existing configurations of the data subscriptions.
- **Remove**: Remove the data subscription from the scenario.

→ Tip

Via `CTRL + Click` you can select multiple data subscriptions in the list at once, as well as edit them simultaneously. This allows you to apply changes (such as a choosing a certain package) to multiple data subscriptions without affecting

the whole list.

5. (Optional) Check assigned packages for the data subscriptions and mass reassign packages.

   It's best practice to assign all data subscriptions in one scenario to the same package as this enables them to be transported together (so you can reconstruct the BW model in other systems as well). Before generating the data subscriptions, you should check whether the assigned packages have any inconsistencies.

   To check and assign packages, right-click on the data subscriptions list and choose:

   - **Check Packages:** Check all packages in the scenario for inconsistencies. The results will be displayed in an informational message.
   - **Assign Package (Changed Only)**: Select a package and mass assign it to all data subscriptions in the list that have been changed for the scenario. This includes data subscriptions that will be newly created.
   - **Assign Package (Local Only)**: Select a package and mass assign it to all data subscriptions in the list that are currently assigned to the local package $TMP. This allows you to mass replace $TMP assignments in one operation.

   In addition, you can assign a new package to an individual data subscription in the list by right-clicking and choosing **Edit** > **Package** > **Browse** >.

6. Once you're finished with all settings, choose **Generate** to generate all required data subscriptions for constructing the scenario.

   The results of the operation will be shown in the **Messages** tab of the Scenario Generator. Here you can see which data subscriptions have been generated successfully, as well as the BW object they correspond with, and their proper technical name. If the creation has failed, you can view the error messages and warnings to understand what went wrong.

Results

The generated data subscriptions for the scenario are now available on your BW system and can be found in the Data Subscriptions view in the BW Modeling Tools (BW Data Product Generator Perspective in BW Modeling Tools (BWMT)), the Data Subscriptions app in the SAP BW/4HANA Cockpit (Data Subscriptions App in SAP BW/4HANA Cockpit), and the RSDPS transaction in the SAP GUI (RSDPS Transaction in SAP GUI).

The scenario data subscriptions can be edited (Editing Data Subscriptions) and deleted (Deleting Data Subscriptions) just like any other data subscriptions.

Unless you have chosen **Activate Directly** or **Activate in Background** as **Mode**, you still need to activate the data subscriptions to make them executable. You can mass activate the data subscriptions in the Data Subscriptions view in the BW Modeling Tools or individually in the SAP GUI and in the SAP BW/4HANA Cockpit.

Generating the data subscriptions doesn't automatically run them as well. You still need to do this manually afterwards to create the local tables (file) and extract the data from the InfoProviders to reconstruct the BW model. The data subscriptions can only be run in the SAP GUI or in the SAP BW/4HANA Cockpit.

Working with Local Tables (File)

Local tables (file) are a table type in SAP Datasphere that is optimized for storing large volumes of data in a cost-effective way using a file-based storage.

The local tables (file) are created and saved in a special file space of type BW in the object store in SAP Datasphere. In contrary to standard spaces in SAP Datasphere, this BW file space is based on storage in SAP HANA Cloud, data lake Files (HDLFS).

ℹ Note

SAP HANA Cloud, data lake Files (HDLFS) is a component of SAP HANA Cloud, data lake. It provides a managed container for storing structured, semi-structured, and unstructured data as files. In the object store of SAP Datasphere it is used to handle

large volumes of data efficiently, allowing big data workloads to run at scale without burdening traditional HANA in-memory storage.

The Data Product Generator directly creates and deploys a local table (file) if you create and save a data subscription. If you run the data subscription, this local table (file) will be populated with data from the selected BW InfoProvider. For more information, see Creating Data Subscriptions.

If you are using multiple targets for the Data Product Generator, you can deploy local table (files) into different BW file spaces in the same SAP Datasphere tenant or even into different SAP Datasphere tenants. The target can be selected when you create a new data subscription. For more information, see Configuring Targets for the Data Product Generator.

In the Data Builder in SAP Datasphere you can view the local table (file), as well as its data and metadata. You can also schedule a merge task to merge new data pushed by the Data Product Generator into the local table (file) after initial extraction. For more information, see Viewing Local Table (File) and Getting Data Updates.

To use the BW data in the local table (file) in models and views in SAP Datasphere, you first need to share it to another space, since the BW file space itself is read-only and doesn't allow to create any objects except for task chains. For more information, see Performing Data Management Tasks and Sharing Local Table (File) to Other Spaces.

Viewing Local Table (File) and Getting Data Updates

You can view the local table (file) including its data and metadata in the Data Builder in SAP Datasphere. The local table (file) can also be directly accessed from the SAP BW/4HANA Cockpit, the SAP GUI, or the BW Modeling Tools. To merge new data pushed by the Data Product Generator into the local table (file), you need to schedule a merge task manually or via task chain.

Prerequisites

- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. During the configuration, the BW administrator has created a space to receive data pushed by the Data Product Generator (BW file space). (For more information, see Configuring the Data Product Generator.)
- The BW administrator has created, saved, and run a data subscription to replicate data from a BW InfoProvider into a local table (file) in the BW file space in SAP Datasphere. As a result, the local table (file) has been successfully generated and deployed in the space. (For more information, see Creating Data Subscriptions.)

> **Note**
> For each data subscription a separate local table (file) will be created. The local table (file) will contain the data extracted from the selected ODP-based source (InfoProvider). This means one InfoProvider maps to one data subscription and through this to one local table (file).

Viewing Data and Fields of Local Table (File)

You can view the read-only data and properties of the local table (file) in your SAP Datasphere tenant, either by logging on to SAP Datasphere and using the Data Builder or by accessing the local table (file) directly from the data subscription.

1. In the Data Subscriptions app in the SAP BW/4HANA Cockpit, you can directly open the local table (file) from within your BW system by selecting **View Local Table (File)** in the **Local Table (File)** column in the data subscriptions list (you might first have to toggle this column via **Settings** (⚙)), or by using the equivalent button in the **Maintain Data Subscription** screen.

2. In the RSDPS transaction in the SAP GUI, you can preview the remote data in the local table (file) by selecting the data subscription and then choosing **Browse** (Browse) in the top bar.

3. In the Data Subscriptions view in the BW Modeling Tools, you can open the local table (file) via `CTRL + Click` on **Show Local Table (File)** in the **Local Table (File)** column in the data subscriptions list.

4. In the Data Builder in SAP Datasphere, you can open a preview of the data in the local table (file) by selecting **View** (View) in the top bar.

   The preview will show the columns of the local table (file) and a select amount of records.

ℹ Note

The data subscription is not only extracting the data out of the InfoProvider, but also some of its metadata (such as fields, data types, ...). This metadata will be carried over into the local table (file), making semantic context from BW objects accessible for further usage.

In the Data Builder, you can also view this semantic context by clicking on the name of the local table (file). Here, you can see its definition including fields, data types, and further properties.

Getting Data Updates for Local Table (File) from BW System

If a data subscription pushes new data or delta data into the same local table (file), the new data will first arrive in the inbound buffer of the object store of SAP Datasphere.

To update the data in the local table (file), you need to manually schedule a merge task or use a task chain in SAP Datasphere.

1. **Manually schedule a merge task:** To merge the data updates manually, you need to create and schedule a Data Task in your SAP Datasphere tenant. Go to the tab **Tasks**, create a new task, select **Merge** as the task type, and configure the source tables (snapshot/delta) and the target tables. Then, schedule the task to run as needed. For more information, see Merge or Optimize Local Table (File).

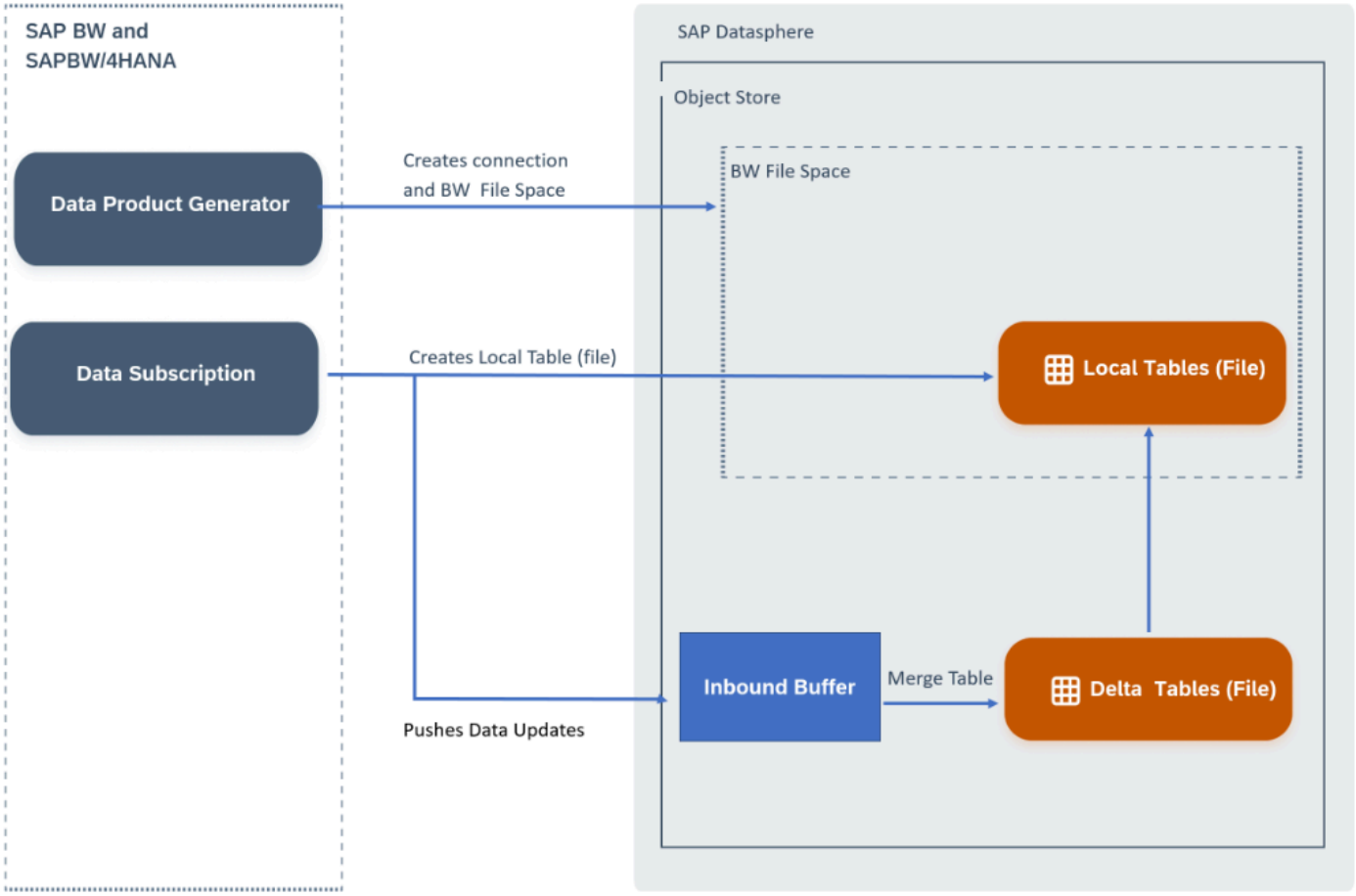
2. **Run a task chain:** To automate this process, you can create a task chain that includes the merge task. Go to the tab **Task Chains**, create a new task chain, add the merge task, and configure a schedule. This way the merge will run automatically according to the defined schedule. For more information, see Creating Task Chain.

ℹ Note

In the inbound buffer, no duplication will happen. The data subscription will continue to push data updates into the inbound buffer in the same table until a merge task has emptied it. For this, it doesn't matter whether the data arrives in several batches at different times (having different timestamps).

→ Tip

If you have created a data product out of the local table (file) via the Data Sharing Cockpit, the data in this data product is updated automatically if the data subscription pushed new data from the BW system and you merge it into the local table (file). There is no manual update for the data product required.



Performing Data Management Tasks and Sharing Local Table (File) to Other Spaces

You can work with the extracted BW data in the local table (file) after you shared it to other spaces. In addition, you can perform data management tasks (such as deleting data) directly in the read-only BW file space. If you access the Data Sharing Cockpit, you can also create a data product out of your BW data.

Prerequisites

- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. During the configuration, the BW administrator has created a space to receive data pushed by the Data Product Generator (BW file space). (For more information, see Configuring the Data Product Generator.)
- The BW administrator has created and run a data subscription to replicate data from a BW InfoProvider into a local table (file) in the BW file space in SAP Datasphere. (For more information, see Creating Data Subscriptions.)

Usage of BW-generated Local Tables (File)

There are some specialties to consider when working with local tables (file) generated by the Data Product Generator for SAP Business Data Cloud:

- The properties of the local table (file) and its structure cannot be changed anymore.
- It's not possible to change settings for delta capture (OFF/ON) anymore. If delta capture is OFF, delta capture changes won't be tracked and instead historical snapshot versions will be kept.
- It's not possible to restore the local table (file) to a previous version. You can only display the previous version.

- If partitions have been defined BW-side at table creation, you can see them, but you can't change them or create new ones.

  ℹ Note

  Partitions can't be maintained for the local table (file), but partition criteria from the associated InfoProvider (e.g. DataStore object (DSO)) can be reused if possible.

- Find and Replace is not available.
- The local table (file) in the BW file space is read-only (as all objects in this space are read-only) - if you want to work with the data, you'll have to share it to another space as a prerequisite.

  → Tip

  If you delete the data subscription within your BW system, the local table (file) becomes editable. For more information, see Deleting Data Subscriptions.

Performing Data Management Tasks for Local Table (File)

While the local table (file) and its data are read-only, you can still perform data management tasks (such as deleting data) on objects in the BW file space.

The following data management tasks are available for your local table (file) in addition to previewing its data:

- **Viewing the Impact and Lineage Analysis diagram of the local table (file)**: You can display the Impact and Lineage Analysis diagram for the local table (file) to analyze its dependent objects (impact) and its data origins (lineage). For more information, see Impact and Lineage Analysis (SAP Datasphere).
- **Monitoring the local table (file)**: You can monitor the local table (file) to check how and when it was last updated and whether new data still needs to be merged. For more information, see Monitoring Local Tables (File) (SAP Datasphere).
- **Exporting the local table (file)**: You can export the local table (file) and its data to a CSN/JSON file. For more information, see Exporting Objects to a CSN/JSON File (SAP Datasphere).
- **Deleting data from the local table (file)**: You can delete records from a local table (file) or versions of it to free up storage space and for housekeeping. For more information, see Deleting Local Table (File) Records (SAP Datasphere) and Deleting Data From Local Tables (File) (SAP Datasphere).

Sharing Local Table (File) to Other Spaces for Consumption in Models and Views

If you want to use the data in the local table (file) to create models and views, you first need to share it to other spaces.

Users in these spaces can then discover and add the shared local table (file) to their space. Once the object has been added to another space, it can be used to create models and views just like any other objects in that space.

You can create models and views in the Data Builder or SQL Editor of SAP Datasphere.

ℹ Note

Sharing is only possible between spaces in the same SAP Datasphere tenant. To be able to share the local table (file), you need a scoped role with the privileges **Data Warehouse Data Builder** (-R----S-) (to share objects to other spaces) and **Spaces Files** (-R------) (to access objects in the space). The **DW Modeler** role template, for example, grants these privileges.

For more information, see Sharing Entities and Task Chains to Other Spaces (SAP Datasphere).

Creating Data Product out of Local Table (File) in Data Sharing Cockpit

If you want to enable your extracted BW data for further use cases within SAP Business Data Cloud, you can use the Data Sharing Cockpit in SAP Datasphere to create a data product out of the local tables (file) in your BW file space.

The data products will appear in the catalog, accessible from both SAP Datasphere and the SAP Business Data Cloud cockpit, and can be shared to SAP Databricks via Delta Sharing where you can execute advanced AI (artificial intelligence) and ML (machine learning) workflows on top of your BW data.

For more information, see Data Provider's Guide - Data Sharing Cockpit (SAP Datasphere) and Creating Data Products (Data Sharing Cockpit).

Creating and Sharing Data Products

Data products are an object type in SAP Datasphere and SAP Business Data Cloud that has been introduced to enable automated data sharing, governance, and consumption across different applications and domains.

You can create data products out of your BW data through the Data Sharing Cockpit in SAP Datasphere:

- Creating Data Products (Data Sharing Cockpit)

The BW data products will be listed in the catalog in SAP Datasphere and the SAP Business Data Cloud cockpit and are from there available for further use cases:

- Sharing Data Products to SAP Databricks

ℹ Note

A data product is a ready to use, curated data set that is delivered for analytics, AI/ML, and operational use cases and provides insights related to a specific business topic or challenge. It consists of a refined data set, metadata describing the data set, and an access port or Application Programming Interface (API), allowing to exchange data with third-party applications.

The BW data product is created directly out of the local table (files) in the BW file space which contains the data extracted from the BW InfoProvider. This is how you can convert, for example, Sales Orders from a DataStore object (advanced) (aDSO) into a BW data product delivering insights about these Sales Orders.

The created BW data products are custom data products (also called custom Delta Share data products or Delta Share data products). This means they are maintained and governed by your own organization and not by SAP.

Creating Data Products (Data Sharing Cockpit)

You can create BW data products out of the local tables (file) in the BW file space by utilizing the Data Sharing Cockpit in SAP Datasphere. The data products will be listed in the catalog in SAP Datasphere and the SAP Business Data Cloud cockpit. From here, they are available for further use cases, such as sharing them to SAP Databricks via Delta Sharing.

Prerequisites

- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. During the configuration, the BW administrator has created a space to receive data pushed by the Data Product Generator (BW file space) and performed all necessary connection setup tasks. (For more information, see Configuring the Data Product Generator.)

- The BW administrator has created and run a data subscription to replicate data from a BW InfoProvider into a local table (file) in the BW file space in SAP Datasphere. (For more information, see Creating Data Subscriptions.)
- The BDC administrator has created a formations profile in the Data Sharing Cockpit in SAP Datasphere. This is a specialized data provider profile that enables the creation and consumption of data products exclusively within a specific SAP system formation (e.g. an SAP Business Data Cloud formation). To configure the formations profile you need to go to **Data Sharing Cockpit** > **My Data Provider Profile** > **Create Data Provider Profile**, fill in the profile, and select the visibility option **Formations**. (For more information, see Creating Custom Delta Share Data Products - Creating Formations Profile (SAP Datasphere).)

Context

To create a data product out of BW data pushed by the Data Product Generator, you need to utilize the Data Sharing Cockpit. This is a dedicated app available in your SAP Datasphere tenant to create and publish data products into the catalog. (For more information, see Creating Custom Delta Share Data Products (SAP Datasphere).

i Note

The created BW data product will be a custom data product on a Delta Share runtime (also called custom Delta Share data product or Delta Share data product). This means it is optimized for sharing to other SAP systems or supported third-party systems such as SAP Databricks via the Delta Sharing protocol.

There are other types of data products that can be created through the Data Sharing Cockpit, which are called marketplace data products. These are proprietary to SAP and published to the catalog in the Data Products (Marketplace) collection in SAP Datasphere. The BW data products however are not marketplace data products.

Procedure

1. In your SAP Datasphere tenant, open the sidebar navigation, and go to **Data Sharing Cockpit** > **My Data Products**.

   Here, you can view a list of all your data products, as well as the space they belong to, their status, their visibility, and their number of installations.

2. To create a new data product, go to the top bar and select **Create Product**.

   A new screen will open where you can fill in the details of the data product.

3. Enter a **Business Name** for your data product (e.g. Sales Orders). The **Technical Name** will be automatically filled in (e.g. SALES_ORDERS).

4. In the section **Overview**, enter a description for your data product (e.g. BW DataStore object (advanced) containing Sales Orders).

5. In the section **Product Details**, select a **Data Category** (e.g. Company Data).

   The field **Contexts** is automatically set to **Formations** since you create the data product as a formations provider. The field **Data Shipments** is also automatically set to **Integrated Delivery** and the field **Delivery Mode** to **Delta Share**. Under **Pricing**, the **Contract Type** is automatically set to **Free**. It is not possible to adjust any of these default values.

6. In the section **Product Artifacts**, select your BW file space as the **Product Artifact Space**.

   The selected space contains the local tables (file) with your BW data. The **Storage Type** will be automatically set to **SAP HANA Cloud, data lake Files** as this is the technology used under the BW file space to store the data.

7. Select the plus icon beside the space name to open a new screen for adding the local table (file) with your BW data. Here, search for and tick off the checkbox for your local table (file). Then, choose **Select** to continue.

8. Select **Save Changes** to confirm your settings and create the data product.

9. In the results screen, select **Switch Status** and choose **List** to list the data product.

   In the results screen the details of the data product will now be filled in and the **Lifecycle Status** will change to **Listing**.

   If you go back to **My Data Products** you can now also see the new data product you just created.

Results

Some time after the data product has been listed, it will be accessible from the catalog in SAP Datasphere and in the SAP Business Data Cloud cockpit under the **Data Products** collection.

> **Note**
>
> It might take a while until the data product appeared in the catalog because the systems need to synchronize first. If you want that it appears immediately, contact your catalog administrator to run a manual synchronization.

Sharing Data Products to SAP Databricks

You can share the BW data products you have created in the Data Sharing Cockpit via Delta Sharing to SAP Databricks in case you have a SAP Databricks capability enabled in your SAP Business Data Cloud formation.

Prerequisites

- The BW administrator has used the Data Sharing Cockpit to create a custom data product on a Delta Share runtime out of the BW data in the local table (file). The data product with the BW data has been successfully listed and appears in the catalog. (For more information, see Creating Data Products (Data Sharing Cockpit).)
- The BDC administrator has enabled SAP Databricks in their SAP Business Data Cloud formation (Provisioning SAP Databricks) or connected their existing Databricks instance as an external system via SAP Business Data Cloud Connect (Provisioning SAP Business Data Cloud Connect).

Context

Sharing the BW data products created in the Data Sharing Cockpit is possible through the catalog entry point in the SAP Business Data Cloud cockpit.

You can share the data products to SAP Databricks in your SAP Business Data Cloud formation, but also to a Databricks instance that is connected to your SAP Business Data Cloud system as an external system through SAP Business Data Cloud Connect.

The share process works through the Delta Sharing protocol, an open protocol used by Databricks for secure data sharing. Delta Sharing allows to share live data across platforms, clouds, and regions without copying or moving it (zero-copy sharing).

Procedure

1. Open the **SAP Business Data Cloud Cockpit** and go to **Catalog & Marketplace** > **Search**.
2. In the catalog, search for the data product you have just created out of the BW data and open its details page.
3. Here, select **Share** to open the **Manage Access** dialog.
4. In the **Manage Access** dialog, you can configure further settings for sharing.
   a. Go to **Target System** and select the SAP Databricks system you want to share to.
   b. Enter a **Share Name** for the data product.

      This is the name the data product will have in the SAP Databricks catalog. You will not be able to change it after sharing.

      > **Note**
      >
      > Only lowercase letters are allowed for the **Share Name**.

   c. Select a **Workspace** in SAP Databricks to share to.

You can select one or multiple workspaces. If there is only one workspace available, it will be automatically selected.

> **Note**
>
> A workspace is an environment in SAP Databricks to access all SAP Databricks assets (such as notebooks, tables, views, SQL queries) for your team or your department in one place. An organization can have multiple workspaces or just one.

d. Once you're finished, select **Share**.

5. A message will appear to let you know the share process has started. Once the process has been completed, the **Manage Access** dialog will refresh and you can choose **Open** to directly open the data product in SAP Databricks for further consumption or enrichment.

   For detailed information on the process, see Sharing Data Products to SAP Systems - Sharing Data Products to SAP Databricks.

   > **Tip**
   >
   > You can also share the data product through SAP Business Data Cloud Connect to an externally connected Databricks system. For more information, see Working with Data Products in SAP Business Data Cloud Connect.

Results

After the data product has been shared to SAP Databricks, the BW data can be used in advanced AI (artificial intelligence) and ML (machine learning) workflows.

- In SAP Databricks, the data product is accessible as a Delta table in the Unity Catalog. The schema and the Delta table can be found in the folder **Delta Share Received**.
- You can act on the data by creating a notebook for tasks like data preparation, data training, data forecasting, and writing the forecast results into a Delta table.
- The Delta table can be shared back again to SAP Business Data Cloud. You can also publish the share as a data product back into the catalog in the SAP Business Data Cloud cockpit.

For more information, see SAP Databricks Documentation .

Automating Tasks and Operations

SAP Business Warehouse, private cloud edition, as well as the Data Product Generator, provide advanced capabilities for the automation of tasks and operations, scheduling of processes, and integration with similar options from SAP Datasphere and SAP Business Data Cloud.

There are two types of scheduling options for processes available:

- **Process chains:** These are automated sequences of processes in SAP BW and SAP BW/4HANA which can be used to manage and control data loading and transformation tasks in a structured way. They link multiple steps (called processes) together and run them in a defined order. This automates ETL (Extract, Transform, Load) operations in BW, such as loading data from source systems, activating DataStore objects (DSOs), running transformations and aggregations, and starting reports or alerts after completion.

  > **Note**
  >
  > The process types are predefined by SAP or can be created by users. There are standard process types available for all activities around data loading and data maintenance as well as some general services (e.g. activate DataStore objects, delete requests, start ABAP program).

The process variants are reusable configurations for the process types to define specific settings applying to them (e.g. in the process type for task chains you can configure which task chain you want to include and in which SAP Datasphere space it is located). The variants correspond with the process steps. You can create multiple variants for the same process type and include them in the same process chain (e.g. for loading data from multiple different sources via the same loading mechanism).

You can find more information on process chains in SAP BW/4HANA (Process Chain (SAP BW/4HANA)) and in SAP BW (Process Chain (SAP BW 7.5)).

- **Task chains:** This is an automation option available in SAP Datasphere to streamline and orchestrate data integration processes. They can be created in the Data Builder in SAP Datasphere by adding different elements to the task chain canvas. As elements you can add, for example, local tables, remote tables and views, data flows, replication flows, transformation flows, and even other task chains.

  ℹ Note

  The BW file space (inbound space) is read-only and it's not possible to create any objects in it, but you can still create task chains. In addition, task chains can be created in other spaces in your SAP Datasphere tenant.

Common use cases for process chains in the Data Product Generator include:

- Schedule the extraction of data to the local tables (file) by running the data subscriptions via process chain and not manually. For more information, see Creating Data Subscriptions.

Common use cases for task chains in the BW file space include:

- Schedule a merge task to automatically merge new data pushed by the data subscription into the local table (file). For more information, see Viewing Local Table (File) and Getting Data Updates.
- Include task chains in process chains from your BW system by configuring a process variant for the equivalent process type in the Process Chain Editor. For more information, see Running Task Chains in Process Chains.

Running Task Chains in Process Chains

You can run task chains from SAP Datasphere within process chains in your SAP Business Warehouse, private cloud edition system by using the process type "Task Chain" that's available from within the Process Chain Editor in the SAP BW/4HANA Cockpit or the Data Warehousing Workbench in the SAP GUI.

Prerequisites

- The BW administrator has set up and configured the Data Product Generator for SAP Business Data Cloud. During the configuration, the BW administrator has created at least one target for the Data Product Generator, as well as performed all necessary connection setup tasks to establish connectivity to SAP Datasphere. (For more information, see Configuring the Data Product Generator and Configuring Targets for the Data Product Generator.)
- The BW administrator has created a task chain in a space in SAP Datasphere using the Data Builder. You can also create task chains in the BW file spaces associated with the Data Product Generator. While the BW file space itself is read-only and you cannot create any objects in it, it's still possible to create task chains. (For more information, see Creating Task Chain (SAP Datasphere).)

Context

Process chains can be created in the Process Chain Editor in the SAP BW/4HANA Cockpit or in the Data Warehousing Workbench in the SAP GUI. If you create a new process chain, you need to define the steps in the process by using process types and process

variants.

With the process type "Task Chain" it's possible to include a task chain from a space in SAP Datasphere in the process chain in your BW system.

Procedure

1. Log on to SAP Datasphere and ensure you have a task chain created in a space for which the user who runs the process chain is authorized.
2. Open the **Process Chain Editor** app in the SAP BW/4HANA Cockpit.

   Here you can view all currently existing process chains in your BW system.
3. Select **Create** to create a new process chain.

   In the process chain canvas that opens you can visually build the process out of the individual process types as building blocks.
4. Select **Add Process** to define a new step in the process chain.
5. Make further settings for the process in the **Properties of Process** dialog.

   a. As **Process Type** select **General Services - Task Chain**.

   b. (Optional) Tick off the checkbox **Variant** if you want to use a process variant that you have previously defined in the Process Variant Editor.

   c. Enter a **Description** for the process (e.g. the name of the task chain).

   d. As **Target** select one of the targets you have configured for the Data Product Generator to establish the connectivity to SAP Datasphere.

   In the value help you can chose from all existing targets you have created for the Data Product Generator, including the target name, description, and target type (HDLFS). If you have a configured a user-preferred target for your BW system, it will be marked up with an X in the column **Preferred**.

   e. As **Space** select the space in which you have created the task chain.

   f. As **Task Chain** select the task chain you have initially created in the space.
6. Save the new process by selecting **Save**.

   Enter a **Description** and a **Name** for the process chain and select the **Package** it should belong to. Choosing an **InfoArea** is optional.

   As a result, you will see the process chain loaded in the Process Chain Editor with the new information.
7. Select **Activate** to make the process chain executable.
8. Then, select **Execute** or **Scheduling** to run the process chain.

   Go to **Monitoring** to view the results of the process chain run or resolve any error messages.

Results

If you run the process chain containing the task chain, not only the process chain will be executed, but also the task chain. This way you can directly run tasks in SAP Datasphere (such as the merge task for the local tables (file)) directly from within your BW system.

Further Resources

Access further links and resources for functions related to the Data Product Generator or for onboarding to SAP Business Data Cloud.

- Onboarding Guide for SAP Business Data Cloud - Get started with SAP Business Data Cloud, set up your system for the first time, and explore core functions and modules such as data products and intelligent applications.

- What's New in SAP Business Data Cloud - Explore latest features released in SAP Business Data Cloud.
- SAP Road Map Explorer (SAP Business Data Cloud) - Explore future features planned for the Data Product Generator and for SAP Business Data Cloud.